

Asymmetric Information in Wage Contracts: Experimental Evidence and Welfare Implications

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Abstract

Compared to self-employment or performance-based pay, fixed wages reduce earnings uncertainty but are prone to market distortions through moral hazard and adverse selection. Using a model of wage contracts under asymmetric information, I show how these distortions can be identified as marginal treatment response functions in a marginal treatment effects (MTE) framework. I apply this framework to a field experiment in which data-entry workers choose between a randomized fixed wage and a piece rate. I find evidence of both moral hazard and adverse selection. Fixed wages reduce worker productivity by an estimated 8.74 percent relative to the mean. Meanwhile, a 10 percent increase in the fixed wage offer attracts a marginal worker whose productivity is higher by 1.90 percent of the mean. Using semi-parametric MTE estimation, I calculate the welfare loss associated with asymmetric information and the marginal values of public funds (MVPFs) for policies aimed at recovering this loss. I find that taxing data-entry workers' piece-rate earnings can mitigate adverse selection into fixed-wage contracts, generating tax revenue at social costs as low as \$0.87 per dollar.

Keywords: compensation structure, wage insurance, performance pay, adverse selection, moral hazard, information asymmetries, marginal treatment effects.

JEL Classifications: J33, J38, M52, D82.

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1 Introduction

Fixed wages offer workers a guaranteed return on each hour of work, even when output is unpredictable. In financial terms, they operate like equity contracts in which the employer buys a claim on the worker’s next hour of labor product. But these contracts are also vulnerable to moral hazard and adverse selection—a fixed wage might induce less effort or attract less productive workers than freelance hiring or piece-rate pay. Such information asymmetry problems can distort equilibrium wages, leaving workers over-reliant on output-based compensation. The potential for these distortions is perhaps highest in short-term gig labor markets, where millions of workers are compensated by the number of miles driven, pages written, or tasks completed. While these workers may benefit from choosing their own hours (Mas and Pallais, 2017; Garin et al., 2020; Koustas, 2018), they also face uncertainty over what they will earn during those hours. Could asymmetric information be to blame?

The potential for market failures surrounding output-based compensation holds important policy implications. Taxes on piece rates, employment classification rules, and even the minimum wage can mitigate the welfare costs of asymmetric information. However, designing these policies requires knowledge of both moral hazard and adverse selection, not just on average, but with respect to marginal wage changes. For example, an optimal piece-rate tax must balance the welfare benefits it provides to the marginal fixed-wage worker with the distortionary costs of encouraging that worker to shirk. Moral hazard and adverse selection are inherent to this trade-off, but identifying their effects on wage contracts is challenging for several reasons. First, both forces can lead to lower observed productivity among fixed-wage workers compared to those with output-based pay, making them hard to distinguish without exogenous variation in wages. Second, estimates that rely on labor supply decisions in the presence of competitive outside options will likely understate the value of fixed wages and overstate supply elasticities relative to their market-level values (Dube et al., 2020; Caldwell and Oehlsen, 2023). Finally, adversely selected contracts may be too unprofitable for firms to offer in equilibrium. Because these “missing contracts”—and the individuals who would choose them—cannot be observed in existing markets, methods that extrapolate from local price variation often exclude the selection margins most relevant to measuring welfare loss (Einav et al., 2010b).

In this paper, I experimentally estimate the equilibrium and welfare effects of

moral hazard and adverse selection in a large online labor market for data-entry work. To identify these forces, I conduct a field experiment with two stages of randomization: First, I offer workers a choice between a randomized fixed hourly wage and a standardized piece rate. Then, after workers choose a payment option but before they begin their assigned task, I increase hourly wages for a randomized subset of those who accepted hourly offers, bringing them to parity with the highest offered wage. Using the initial wage offer as an instrument for accepting an hourly contract allows me to identify moral hazard as the treatment effect of fixed-wage compensation on worker output. Meanwhile, comparing output across workers on the same contract who faced different ex-ante offers identifies adverse selection into fixed wages.

To map my experimental estimands into a welfare framework, I show how both moral hazard and adverse selection can be expressed using marginal treatment response (MTR) functions under fixed-wage and piece-rate counterfactuals—the objects identified in a marginal treatment effects (MTE) analysis (Björklund and Moffitt, 1987; Heckman and Vytlačil, 1999, 2005, 2007). Embedding these MTR functions into a model of asymmetric information allows me to flexibly characterize equilibrium and efficient allocations of fixed-wage labor. Much like canonical models of insurance markets (Einav et al., 2010a; Akerlof, 1970), equilibrium is determined by two curves: a worker’s hourly reservation wage—the minimum payment they will accept for an hour of labor—and the average value of output among workers with lower reservation wages than their own. A worker cannot be profitably hired for a fixed-wage job if their reservation wage exceeds the average output value of lower-reservation-wage workers. Relative to a full-information equilibrium, this profit condition leads to an underprovision of hourly work—some piece-rate workers would like to forfeit a portion of their expected earnings in exchange for a fixed-wage contract, but the threat of adverse selection prevents employers from offering such contracts at those workers’ reservation wages. My empirical design allows me to estimate the welfare loss implied by these distortions using semi-parametric MTE methods.

Results from my experiment provide evidence of both moral hazard and adverse selection into fixed-wage contracts. Estimated treatment effects imply that working under hourly pay reduces workers’ output value by 8.74 percent relative to the mean. At the same time, a ten-percent increase in the hourly wage offer attracts a marginal worker with 1.90-percent higher productivity compared to the mean. This finding appears driven by selection on *unobserved* determinants of productivity; re-

sults persist after conditioning on demographics and work experience, suggesting that pre-employment screening cannot fully resolve asymmetric information problems in this labor market.

Using local polynomial MTE estimation (Carneiro et al., 2011), I find that 58 percent (SE=1.88) of workers in my sample would pay a premium for fixed-wage contracts over piece-rate pay. This share reflects the efficient allocation that would exist if employers were fully informed of workers’ potential output. By contrast, only 49 percent (SE=1.56) of workers would find hourly positions in a competitive equilibrium with adverse selection and moral hazard. These two equilibria reflect a non-linear pattern of selection on moral hazard: At low reservation wages, those with marginally stronger preferences for fixed wages have a slightly higher propensity to shirk, consistent with patterns of selection in insurance markets (Einav et al., 2013). At higher reservation wages, however, shirkers are more likely to prefer piece rates, a pattern that could be explained by heterogeneous risk preferences—those who prefer the variability of piece-rate pay may also be more likely to shirk under fixed wages because they are less fearful of poor performance reviews.

Estimated MTR functions imply that the welfare loss arising from this inefficient equilibrium is \$0.05 (SE=0.017) per hour of labor. Scaling this per-hour loss by roughly 35 billion annual paid worker-hours in online clerical and data-entry gig work implies aggregate annual welfare losses on the order of \$1.7 billion globally. Using an MVPF analysis (Hendren and Sprung-Keyser, 2020), I show how governments can efficiently recover these losses with policies that balance the welfare benefits of promoting fixed wages with the fiscal costs of increased shirking. I find that a tax on piece-rate earnings can yield a marginal cost of public funds as low as 0.87, meaning each dollar of tax revenue carries a net social cost of just \$0.87—more efficient than a distortionless tax.

These estimates are specific to the data-entry workers in my experimental setting; as in much of the applied literature on worker incentives, both selection and treatment effects may not extrapolate across labor markets. In the final section, I discuss external validity and show how my framework can be applied to other settings. To illustrate, I calibrate my model to the market for rideshare drivers using experimental estimates from Angrist et al. (2021). I find that adverse selection into less ride-dependent compensation results in an approximate welfare loss of \$0.09 per worker-hour, implying a marginal social cost of just \$0.86 per dollar of piece-rate

taxes. This exercise illustrates how my methods can be used to inform policies around tips, commissions, and bonuses in a variety of labor markets.

This study relates to several strands of labor research. Theory shows how asymmetric information can generate shirking, sorting, and inefficient wage setting or labor allocation (Mirrlees, 1971; Miyazaki, 1977; Holmström, 1979; Grossman and Hart, 1983; Jovanovic, 1982; Greenwald, 1986; Lazear, 1986; MacLeod and Malcomson, 1989; Levine, 1991; Kugler and Saint-Paul, 2004; Moen and Rosen, 2005; Shimer, 2005; Stantcheva, 2014).¹ Empirical studies document incentive and sorting effects across compensation schemes in windshield repair (Lazear, 2000), teaching (Brown and Andrabi, 2021; Johnston, 2025), rideshare (Angrist et al., 2021; Caldwell and Oehlsen, 2023), salons (Butschek et al., 2022), mining (Shearer, 1996), health care (Kantarevic and Kralj, 2016; Gaynor et al., 2004), and agriculture (Foster and Rosenzweig, 1996, 2022).² Experimental evidence on selection and treatment effects of payment schemes includes Dohmen and Falk (2011), DellaVigna and Pope (2018), Shearer (2004), and Guiteras and Jack (2018).

This paper also contributes to research on insurance and insurance-like contracts. The model draws directly on Einav et al. (2010a) and Herbst and Hendren (2024), and the empirical mapping to marginal treatment effects complements Kowalski (2023b). I extend this line of work by using MTE methods to quantify welfare loss from adverse selection and moral hazard in fixed-wage labor contracts.

Methodologically, my experimental design builds upon Karlan and Zinman (2009), who randomize contract offerings on microfinance loans in South Africa. Similar to the second stage of my experiment, they isolate selection on unobservables by comparing borrowers who received different initial offers but ultimately faced the same contract terms. They find strong evidence of moral hazard and weaker evidence of adverse selection. In a related experiment, Bryan et al. (2015) estimate how referral bonuses induce selection into consumer loans.³ They estimate large treatment effects of social

¹Related work links information asymmetries to “efficiency wages”—wages paid above the market-clearing rate—(Weiss, 1980; Krueger and Summers, 1988; Weiss, 2014; Yellen, 1984; Malcomson, 1981; Katz, 1986).

²Recent evidence also points to both productivity and selection effects in other aspects of labor contracts like remote work (Emanuel and Harrington, 2024).

³Like my study, they identify selection on potential outcomes (repayment propensity) with and without treatment (peer enforcement bonuses). But unlike my marginal treatment effects approach, which relies on observed outcomes among those who opt out of treatment, their design randomly removes the treatment condition for a subset of those who select into it.

pressure on repayment, but find little evidence of selection on potential repayment or resistance to social pressure.

This study makes three contributions to the existing literature. First, I provide new evidence on how moral hazard and adverse selection can lead to an underprovision of fixed-wage jobs. While previous work has documented the presence of these forces in various labor markets, I develop an insurance-based model of wage contracts to demonstrate their effects on equilibrium and worker welfare. My experiment serves to both quantify these effects in a specific work setting and validate my model more generally, highlighting the potential for suboptimal wage contracts in the broader labor market. Thus, my findings reveal an important and unexplored channel through which many labor policies could improve workers' well-being.

Second, this paper investigates the public costs and benefits of policies aimed at reducing workers' earning risk. Specifically, I estimate the marginal cost of public funds (MCPFs) from taxes on piece rates paid to online gig workers. The components of these MCPFs map directly to my experimental estimands, allowing me to flexibly compare the welfare benefits of inducing more fixed wages with the fiscal costs of increased worker shirking. Aside from its direct relevance to the regulation of online labor platforms, this analysis may shed some light on potential policy trade-offs of taxing tips, commissions, or bonuses in other labor markets.

Finally, this paper provides a methodological framework to flexibly estimate welfare loss from information asymmetries in markets for insurance or insurance-like contracts. Building on methods from Karlan and Zinman (2009) and Bryan et al. (2015), my approach separately identifies treatment and selection over a range of experimental contract offers. Importantly, these offers include contracts that may not be profitable to a real-world firm, avoiding the “under-the-lamppost” problem inherent to many empirical studies of information asymmetries (Einav et al., 2010b). Moreover, because I observe outcomes for both accepters and decliners of these contracts, I can use MTE methods to identify marginal selection on potential outcomes in both insured and uninsured states. The resulting MTR functions map directly to the components of a “cost-curve” insurance model (Einav et al., 2010a; Herbst and Hendren, 2024), allowing me to semi-parametrically identify welfare loss from asymmetric information. Applying this flexible estimation approach to a wide range of experimental contract offers improves upon traditional methods that rely on local price changes and linear extrapolation (Einav et al., 2010a; Hackmann et al., 2012,

2015; Finkelstein et al., 2019).

The rest of this paper proceeds as follows: In Section 2, I describe my experiment and underlying empirical strategy. In Section 3, I discuss the baseline results of the experiment. Section 4 presents a model of hourly wage contracts under asymmetric information, and Section 5 estimates that model using MTE methods. Section 6 uses estimated MTR functions and associated value curves to calculate MVPFs for fixed-wage subsidies and piece-rate taxes. In Section 7, I discuss the broader implications of my findings and calibrate my model to the market for rideshare drivers. Section 8 concludes.

2 Experimental Design

In this section, I describe my experimental design and empirical strategy. The goal of my experiment is twofold: First, I want to identify the incentive effects of fixed wage contracts on worker performance (moral hazard). Second, I want to identify how workers with different unobserved productive potentials self-select into these contracts (adverse selection). Separately identifying these forces poses an empirical challenge—differences in realized output between workers who opted into a given wage offer reflect both the ex-ante productivity differences between those self-selected groups and the causal effect of the different wage offers they chose.

To overcome this challenge, my experiment offers data-entry workers a choice between a randomized fixed hourly wage and a standardized piece rate.⁴ Comparing realized output between individuals who faced different hourly wage offers but ultimately work under the same contract identifies adverse selection—both groups ultimately face the same compensation scheme but made decisions under different menus of options. At the same time, using wage offers as an instrument for take-up of the hourly contract allows me to separately identify treatment effects of hourly wages among those who accept the offer. I first illustrate the intuition behind this design with a stylized two-contract setting before extrapolating to settings with multiple wage offers.

⁴My design holds piece rates constant because employers face no costs of adverse selection or moral hazard from varying the piece rate—they simply want to pay the lowest possible price per unit of labor product. In other words, the experimental piece rate represents the “no-insurance” outside option to potential fixed-wage contracts. See footnote 23.

2.1 Example using a Single Wage Offer

Consider a potential outcomes framework in which a worker i chooses one of two mutually exclusive contracts—a piece rate and an hourly wage. Let Y_{1i} denote i 's output if they work under the hourly wage, and let Y_{0i} denote their output if they work under the piece rate. Given these potential outcomes, worker i 's observed output, Y_i , is given by $Y_i = D_i Y_{1i} + (1 - D_i) Y_{0i}$, where D_i is a binary indicator for whether i chooses the hourly wage. Differencing realized outputs between hourly workers ($D_i = 1$) and piece-rate workers ($D_i = 0$) would yield the sum of two components—average treatment-on-the-treated and average selection on untreated outcomes:

$$\begin{aligned} E[Y_i|D_i = 1] - E[Y_i|D_i = 0] \\ = \underbrace{E[Y_{1i} - Y_{0i}|D_i = 1]}_{\text{Average Treatment on the Treated}} + \underbrace{E[Y_{0i}|D_i = 1] - E[Y_{0i}|D_i = 0]}_{\text{Average Selection on } Y_0}. \end{aligned} \quad (1)$$

The treatment-on-the-treated component corresponds to the average effect of hourly pay ($Y_{1i} - Y_{0i}$) among those who accept the hourly wage offer ($D_i = 1$), while the selection component corresponds to the average difference in potential piece-rate outcomes (Y_{0i}) between those choosing hourly pay ($D_i = 1$) and those choosing the piece rate ($D_i = 0$). These components are difficult to separate because piece-rate outcomes among hourly workers ($Y_{0i}|D_i = 1$) are not observed.

To separate these forces of treatment and selection, my experimental design randomizes the menu of compensation options faced by workers. In the simple two-contract setting presented above, that means randomly assigning each worker to one of two offer conditions, $W_i \in \{0, 1\}$. Only workers assigned to $W_i = 1$ are offered the choice between the piece rate ($D_i = 0$) and hourly wage ($D_i = 1$), while workers assigned to $W_i = 0$ are paid the piece rate with no alternative. Assume the offer condition W_i can only affect Y_i through the choice of contract, so $W_i \perp\!\!\!\perp (Y_{1i}, Y_{0i})$. Finally, let D_i^* denote worker i 's potential take-up of the hourly wage if given the option ($W_i = 1$), so observed take-up D_i is given by $D_i = W_i D_i^*$.

Comparing worker output across these two treatment-offer groups and scaling by the hourly-wage take-up rate yields the classic treatment-on-the-treated estimator

from Wald (1940):

$$\underbrace{E[Y_{1i} - Y_{0i} | D_i^* = 1]}_{\text{Average Treatment on the Treated}} = \frac{E[Y_i | W_i = 1] - E[Y_i | W_i = 0]}{\pi}, \quad (2)$$

where $\pi \equiv Pr(D_i = 1 | W_i = 1)$, the share of hourly contracts accepted among those offered a choice ($W_i = 1$). In the context of this paper, however, the selection component from Equation (1) is equally as important as treatment effects. I can identify this component by simply comparing output between piece-rate workers in the control group ($W_i = 0$) and piece-rate workers in the hourly-offer group ($W_i = 1$), who declined the hourly wage offer:

$$\underbrace{(E[Y_{0i} | D_i^* = 1] - E[Y_{0i} | D_i^* = 0])}_{\text{Average Selection on } Y_0} = \frac{E[Y_i | W_i = 0] - E[Y_i | D_i = 0, W_i = 1]}{\pi}, \quad (3)$$

where equality follows from randomized assignment.⁵

Figure 1, Panel A illustrates the intuition from Equation (3). The control group, by construction, is subject to the standardized piece rate, while the treatment-offer group is offered an hourly wage as an alternative. Selection is identified by comparing the control group (top left box) to those in the treatment group who chose to remain on the piece rate (top right box). This selection-on-unobservables estimator captures the average difference in potential untreated outcomes for “compliers” versus “never-takers” (Black et al., 2022; Kowalski, 2023a; Mogstad et al., 2018; Huber, 2013).

2.2 Multiple Wage Offers and Second-Stage Randomization

The example above provides a simplified illustration of my experimental design with a binary treatment assignment, $W_i \in \{0, 1\}$. In practice, however, my experiment features several treatment groups facing different hourly wage offers. Including multiple wage offers with incomplete take-up allows me to identify selection on potential outcomes under both the piece rate (the untreated state, Y_0) and hourly wages (the

⁵Randomized assignment implies $E[Y_{0i} | W_i = 1] = E[Y_{0i} | W_i = 0] = E[Y_i | W_i = 0]$, so $E[Y_{0i} | D_i = 1, W_i = 1] = \frac{E[Y_i | W_i = 0] - (1 - \pi)E[Y_i | D_i = 0, W_i = 1]}{\pi}$. Equation (3) can also be derived by subtracting the Wald estimator (2) from the difference in hourly versus piece-rate outcomes in the treatment-offer group, $E[Y_i | D_i = 1, W_i = 1] - E[Y_i | D_i = 0, W_i = 1]$.

treated state, Y_1) across multiple margins.⁶ To see how, consider an example experiment with three offer conditions, $W_i \in \{0, L, H\}$. As in the previous example, control workers assigned to $W_i = 0$ are offered the piece rate with no alternative. But now the remaining workers are randomly separated into two groups—workers assigned to $W_i = L$ are offered the choice between the piece rate and low hourly wage, while workers assigned to $W_i = H$ are offered the choice between the piece rate and a high hourly wage.

As before, I can identify selection on Y_0 by comparing outcomes between piece-rate workers who faced different ex-ante hourly offers—none ($W_i = 0$), low ($W_i = L$), or high ($W_i = H$). Because all three of these groups face the same ex-post payment terms but different ex-ante wage offers, comparisons between them isolate worker selection on potential output under the piece rate, Y_0 . But now I can also identify selection on Y_1 , potential output under hourly pay, by comparing between hourly workers who *accepted* their respective wage offers.

Importantly, however, my identifying assumption requires wage offers to only affect workers’ output through their choice of hourly versus piece-rate contract, $W_i \perp\!\!\!\perp (Y_{1i}, Y_{0i})$. If accepters in both high- and low-offer groups work under the wages they were offered, this exclusion restriction could be violated through wage effects—higher pay might induce greater effort through increased motivation or satisfaction, biasing my estimates. To separate these wage effects from the incentive effects of hourly contract structure, the second stage of my experiment increases wages for a random subset of workers accepting the low hourly wage offer, equalizing their pay with the high hourly wage offer just before the task begins. This surprise top-up creates random variation in *effective* wages among those accepting a given *offered* wage, allowing me to separate potential wage effects from moral hazard and adverse selection.⁷

The intuition behind the multiple wage-offer experimental design is illustrated in Figure 1, Panel B. The top row of boxes represents individuals in each of the three experimental groups, $W_i \in \{0, L, H\}$, who remain on the piece rate. Because all three of these groups face the same ex-post payment terms but different ex-ante wage

⁶Details are provided in Appendix B.1.

⁷Because variation between offered and effective wages is generated through unexpected top-ups, estimated wage effects also include potential behavioral responses to receiving a “surprise” pay raise. As I discuss in Section 3, the wide range of wage offers in my design allows me to separate such responses from true wage effects—a response to surprises should persist for very small wage top-ups, whereas wage effects should reduce to zero as the effective wage approaches the offered wage.

offers, comparisons between them isolate worker selection on productivity under the piece rate, Y_0 . The bottom two boxes represent workers who accepted low ($W_i = L$) and high ($W_i = H$) hourly wage offers, respectively. In the second stage of the experiment, a random subset of those accepting the low hourly wage are promised an additional top-up compensation before they begin working on the task. This surprise top-up equalizes their effective wage with that of the high-offer group, allowing me to separate wage effects (diagonal arrow) from selection on productivity under hourly wages, Y_1 (horizontal arrow).

2.3 Experimental Setting and Implementation

The design and recruitment details for this experiment were pre-registered on the AEA RCT Registry under ID AEARCTR-0000714 (Herbst, 2024). Appendix C provides complete details on the experimental protocol.

Participants in my experiment were recruited on Prolific, an online platform that allows clients to hire online workers for short-term tasks.⁸ The experimental job posting offered workers a \$1.00 reward plus a \$0.03 bonus for each correctly typed sentence transcribed over five minutes. Such transcription tasks are commonly requested on Prolific and other online platforms, often for the purpose of training artificial intelligence (AI) algorithms. Participants were not informed that they were part of an experiment until after they performed the task.

Upon accepting the experimental job, workers were randomized into one of eighteen experimental groups. Each group was offered a different menu of compensation options in exchange for completing the five-minute data-entry task. In the first treatment group, workers were offered a choice between a fixed \$0.10 payment (\$1.20 per hour) or a piece rate of \$0.03 per correctly typed sentence.⁹ In the second treatment group, workers were offered a choice between a fixed \$0.15 payment (\$1.80 per hour) or the same \$0.03 piece rate. Additional treatment groups follow the same structure, with each condition offering the \$0.03 piece rate but increasing the flat wage offer by multiples of \$0.05, up to a maximum of \$1.75 (\$21.00 per hour). A control group was offered the \$0.03 piece rate for each correctly typed sentence, with no alternative op-

⁸Douglas et al. (2023) finds that the Prolific platform compares favorably to Amazon Mechanical Turk (“MTurk”) and other platforms across several dimensions of data quality.

⁹A piece rate of \$0.03 per correctly typed sentence was chosen to roughly align with the market rate for online text-to-text transcription services (GMR Transcription, 2024; GoTranscript, 2024; Ditto Transcripts, 2024; Transcription Services, 2024).

tion. Each of these options was offered as an addition to the \$1.00 reward advertised in the job posting, which all workers received for agreeing to the task. Experimental conditions are listed in Table 1.

The experiment took place in ten waves of three-hundred job postings. Waves were launched at a broad range of days and times to ensure the sample was not restricted to workers with a particular schedule. Data on task performance was collected at the conclusion of each wave. The primary outcome of interest is hourly output value, defined as

$$\text{Output Value} \equiv \frac{\text{Correct Sentences} \times \$0.03}{1/12 \text{ Hours}}. \quad (4)$$

Output value is linked to self-reported background information from workers’ Prolific profiles. Specifically, I observe each workers’ gender, ethnicity, age, employment status, and whether they are a student. I also observe the prior number of tasks they have successfully completed through the Prolific platform. Because the goal of my experiment is to identify selection on *private* information, conditioning on these potentially screenable characteristics is important to simulate a sample of workers who would be observably equivalent to a hypothetical employer.

Importantly, clients on the Prolific platform have the ability to reject or approve a given worker’s assignment. Rejected assignments do not earn rewards and lower workers’ approval ratings. The reputational damage from rejected assignments is a salient concern among workers on Prolific and similar platforms.¹⁰ As in most labor markets, this threat of rejection creates an incentive for online workers to maintain a minimum standard of performance, even if they are paid a fixed wage.

2.4 Why Online Data-Entry?

The online labor platform and data-entry job used in my experiment offer several distinct advantages over alternative settings or tasks. First, online labor markets and data-transcription services represent a large and growing part of the broader gig economy. Over 160 million workers are registered on online labor platforms (Kässi and Lehdonvirta, 2018; Kässi et al., 2021), which have drawn increased attention from both policymakers and academics (Stanton and Thomas, 2025; Horton, 2025; Caplin et al., 2024; Pallais and Sands, 2016; Pallais, 2014; Horton, 2017; Horton

¹⁰See, for example, the Reddit forum r/ProlificAc (<https://www.reddit.com/r/ProlificAc/comments/11yt72u/>).

et al., 2011). Meanwhile, the market for AI “data-labeling” tasks like the one in my experiment has been valued as high as \$19 billion and is expected to grow rapidly (Grand View Research, 2024; Mordor Intelligence, 2025). Second, my experimental task provides a measure of worker output that directly maps to the employer’s profit function and is observable for both accepters and decliners of a given wage offer. Third, I can control the complete menus of wage contracts, so I can observe worker selection between a single predetermined outside option (the \$0.03 piece rate) and a wide range of experimental wage offers, including those that would be unprofitable to a real-world employer. These margins of selection are critical determinants of welfare loss but impossible to observe in most settings.¹¹ Fourth, I can recruit a representative sample of workers in the targeted labor market, not just those opting into a particular employer or wage contract. By contrast, an experiment conducted at an existing workplace would likely exclude workers who prefer self-employment or freelance work, eliminating the very margin of selection I seek to identify. Finally, the protocol for this experiment can be easily replicated: Researchers can validate or extend my experimental intervention in comparable populations of online workers at minimal cost.

3 Baseline Experimental Results

This section describes baseline reduced-form results from the experiment. A total of 3,030 workers participated in the experiment: 302 workers were assigned the control condition with no fixed wage offer, and the remaining 2,728 workers were each assigned to one the 17 wage-offer conditions.¹² Sample sizes for each experimental

¹¹For example, an experiment that recruits participants through randomized posted wages can only measure selection relative to workers’ existing outside options, which likely include competing offers from other fixed-wage employers (Dube et al., 2020). Estimates of selection among these workers would likely understate the insurance value of fixed wages and overstate the elasticity of fixed-wage labor supply. By contrast, my design ensures that participants’ choice sets only include experimental offers, which they observe after accepting a generous upfront fee recruiting them to the task.

¹²One worker exited the task before observing their experimental wage offer and was dropped from the experimental sample. All other workers remained in the sample, even if they failed to enter sentences or click the submit button after the five-minute timer expired. If a worker failed to click the submit button within thirty minutes of accepting the task, the Prolific task scheduler automatically re-assigned their treatment condition to a new worker, even while the unsubmitted task remained in my sample. These re-assigned treatments result in an observation count ($N = 3,030$) that exceeds my pre-registered sample size of 3,000.

group are listed in Table 1.¹³

Table 2 reports summary statistics for the experimental sample. Excluding those assigned to the no-offer control group, 49 percent of workers accepted their experimental fixed-wage offers. Across all experimental conditions, workers completed an average of 21.98 sentences within five minutes, of which 19.45 were typed correctly, resulting in a mean hourly output value of \$7.00.

Hourly Labor Supply The bar chart in Figure 2 shows the share of workers in each treatment group who accepted their hourly wage offer instead of the \$0.03 piece rate. Unsurprisingly, the relative supply of hourly workers increases with the offered wage. On average, wage offers of \$3 or lower were accepted at a rate of 21 percent, while wage offers above \$10 were accepted at a rate of 74 percent. To fit these results to a continuous supply curve, Table 3 reports estimated coefficients from a logistic regression of a binary indicator for hourly acceptance against log wage offer, excluding the control group. Column 1 reports estimates from a univariate specification, while Columns 2 through 4 successively add controls for task timing, employment, and demographics. In each specification, I find a statistically significant effect of log wage offer on hourly take-up, with estimates ranging from 1.20 (SE=0.06) to 1.25 (SE=0.06) depending on the inclusion of controls.

3.1 Evidence of Moral Hazard and Adverse Selection

Figure 2 illustrates how gradual increases in experimental wage offers result in incrementally higher rates of take-up, creating different groups of accepters (hourly workers) and decliners (piece-rate workers) for each hourly offer. In Figure 3, I examine how output value for these groups varies with the generosity of the wage offer.¹⁴ For readability, experimental conditions are aggregated into four categories—those in the control group who received no hourly offers, those receiving wage offers between \$1.20 and \$3.00, those receiving wage offers between \$3.60 and \$9.60, and those receiving wage offers of \$10.80 or higher. Blue circles measure average output values among all individuals in each of these categories. Orange bars measure average out-

¹³Appendix Table A1 reports balance tests of joint significance between baseline demographics and group assignment.

¹⁴Appendix Tables A3 and A4 report the corresponding treatment and selection estimates when output value is instead defined using completed sentences, regardless of correctness.

put values among workers on the piece rate. Dark green bars measure average output values among workers who accepted their hourly wage offers and received randomized top-ups, bringing their effective hourly wages to the \$21.00 per hour maximum. Light green bars measure average output values among those who accepted their hourly wage offers and did not receive top-ups. Following Section 2, I compare these different groups across wage-offer categories to identify treatment and selection into hourly pay, providing evidence for both moral hazard and adverse selection.

Treatment Effect of Hourly Wages In the absence of wage effects, comparing means across each aggregated group in Figure 3 identifies the intent-to-treat effect of hourly wages. The blue circles decline with the generosity of the wage offer, suggesting hourly wages reduce worker output, consistent with moral hazard. Those in the control group, who received no hourly wage offer, produce \$7.28 (SE=\$0.18) of output value. Those receiving wage offers between \$1.20 and \$3.00 produce \$7.11 (SE=\$0.11) of output value, those receiving offers between \$3.60 and \$9.60 produce \$6.98 (SE=\$0.09) of output value, and those receiving offers of \$10.80 and above produce \$6.84 (SE=\$0.10) of output value.

To obtain point estimates for the treatment effect of hourly wages on workers' output value, I disaggregate the groups from Figure 3 into a continuous measure of wage offers, which I use as an instrumental variable for hourly contract take-up. To remove the potential influence of wage effects, I first regress output value against log effective hourly wages (i.e., inclusive of top-ups) and a full set of experimental group dummies for the sample of hourly workers who were eligible to receive random top-ups. I then residualize hourly workers' output values by subtracting the demeaned wage effect implied by the coefficient on log effective hourly wages. I use this residualized measure of output value as the dependent variable in a two-stage least-squares (2SLS) regression where I instrument for hourly wage take-up with log wage offers and an indicator variable for being in the no-offer control group.¹⁵

Table 4 reports 2SLS estimates of the treatment effects of hourly wages on output value. As in Table 3, Column 1 reports estimates from a univariate specification, while Columns 2 through 4 successively add controls for task timing, employment, and demographics. Across all four specifications, hourly contracts reduce worker pro-

¹⁵Appendix Table A2 reports estimates from 2SLS regressions using unresidualized output values. Results from these specifications are nearly identical to baseline results that adjust for potential wage effects.

ductivity; the estimates are significant at the 1 percent level in Columns 1 through 3 and at the 5 percent level in Column 4. Absent controls, accepting an hourly contract reduces a worker’s output value by \$0.61 (SE=0.22) or 8.74 percent of the sample mean. Adding controls for task specifics changes this estimate to \$0.61 (SE=0.21), while adding employment and demographic controls reduces it to \$0.59 (SE=0.21) and \$0.48 (SE=0.19), respectively. These negative and significant treatment effect estimates are consistent with moral hazard—on average, workers’ output is lower under hourly wages, when they do not bear the financial cost of decreased effort, than under piece rates, when their earnings are more closely tied to output.

Selection into Hourly Wages While comparisons of pooled means across wage-offer groups identifies treatment effects, comparisons of means for accepters and decliners across wage-offer groups identifies selection on potential output under hourly wages and piece rates, respectively. In Figure 3, the orange bars increase with the wage offer, meaning those who decline the most generous hourly wages in favor of the piece rate have relatively high output. Those declining offers between \$1.20 and \$3.00 produce \$7.66 (SE=\$0.12) of average output value, those declining offers between \$3.60 and \$9.60 produce \$7.87 (SE=\$0.13) of average output value, and those declining offers of \$10.80 and above produce \$7.84 (SE=\$0.23) of average output value. All three of these averages exceed the \$7.28 (SE=\$0.18) of average output value produced by exclusively piece-rate workers in the no-offer control group. These patterns are consistent with adverse selection on Y_0 , potential output under the piece rate.

The green bars also rise with the wage offer, meaning those who accept the lowest hourly offers over the piece rate have relatively low output. Restricting attention to top-up workers who were paid the same effective rate of \$21.00 per hour (dark green bars), I find that those accepting offers between \$1.20 and \$3.00 produce \$4.62 of average output value, those accepting offers between \$3.60 and \$9.60 produce \$6.04 of average output value, and those accepting offers of \$10.80 and above produce \$6.56 of average output value. These patterns are consistent with adverse selection on Y_1 , potential output under the hourly wage. Note that the average output values among hourly workers who did not receive wage top-ups (light green bars) exhibit a similar pattern to topped-up workers’ averages, suggesting that wage effects are not important in this setting.

Disaggregating the groups from Figure 3 into respective experimental wage offers, I use ordinary-least squares (OLS) estimation to fit the selection patterns seen in Figure 3 to the following linear model:

$$Y_i = \alpha D_i + \beta_0(1 - D_i) \times W_i + \beta_1 D_i \times W_i + \gamma D_i \times W_i^P + \boldsymbol{\xi} \mathbf{X}_i + \epsilon_i, \quad (5)$$

where W_i is worker i 's log hourly wage offer, D_i is a binary indicator for whether they accept the hourly wage, W_i^P is the log wage hourly workers are actually paid (equal to zero for piece-rate workers), and $\mathbf{X}_i$ represents a vector of covariates, a constant term, and a dummy variable for being in the control group.¹⁶

Table 5 reports OLS estimates of coefficients from Equation (5). In Column 1, the estimated coefficient on “Declined $\times$ Log Hourly Wage Offer” implies that increasing wage offers by one log point corresponds to a \$0.09 (SE=\$0.10) increase in output value among those declining the offer in favor of the piece rate, while the coefficient on “Accepted $\times$ Log Hourly Wage Offer” implies that productivity among hourly workers increases by \$0.67 (SE=\$0.12) per log point. The latter coefficient is significant at the 1 percent level in all four columns. The former remains positive in all four columns but reaches conventional significance only in Column 4, where it is significant at the 10 percent level. By comparison, the estimated coefficient on “Accepted $\times$ Log Effective Hourly Wage” is small and statistically insignificant, once again implying that wage effects are not important in this setting. Adding controls for task experience, employment, and demographics in Columns 2 through 4 produces estimates that are more precise and similar in magnitude, suggesting worker selection on ex-ante productivity is not captured by these observable characteristics.

Figure 4 plots OLS estimates from a modified version of Equation (5) that replaces the linear wage-offer term, W_i , with a full set of dummy variables for each experimental wage offer. Covariates include log effective wages among hourly workers and task timing. Orange dots represent coefficients on hourly wage offers interacted with an indicator for remaining on the piece rate. Green diamonds represent coefficients on hourly wage offers interacted with an indicator for accepting the offer. The upward

¹⁶Rather than include a common W_i term and only one interaction term, Equation (5) includes full interactions of wage offers with acceptance status, $(1 - D_i) \times W_i$ and $D_i \times W_i$. While the two models are equivalent, the parameterization of β_0 and β_1 in the fully interacted specification is easier to interpret. Note that $(1 - D_i) \times W_i^P$ is excluded because $W_i^P = 0$ for all $D_i = 0$ (piece-rate workers are ineligible for top-ups).

slope in both of the two series indicates adverse selection into hourly wages—as wage offers decrease, the most productive workers opt out of hourly work and into the piece rate, resulting in lower average productivity among both hourly and piece-rate workers.

Instrument Validity In the description above, I interpret my experimental estimates as treatment and selection effects of fixed-wage contracts. In theory, this interpretation could be threatened by behavioral responses to the experimental intervention. For example, workers’ effort might exhibit reference dependence after being “primed” with a wage offer before the task begins.¹⁷ If workers place a premium on earning at least as much as they were offered, their potential piece-rate output, Y_0 , might increase with the generosity of their experimental wage offer, W_i , violating the exclusion restriction. In this case, however, we would expect piece-rate workers’ output to discontinuously decline when their earnings meet the wage-offer reference point, $Y_i \geq W_i$ (McCrary, 2008).¹⁸ Appendix Figure A1 reports results from a partial test of this hypothesis and finds no evidence of such discontinuity. Using methods from Cattaneo et al. (2020), I estimate piece-rate earnings densities on either side of the wage-offer threshold and find no significant difference (p-value=0.803), suggesting minimal influence from reference points.

A related concern is the potential presence of menu or framing effects (Simonson and Tversky, 1992). If workers’ productivity was influenced not only by their compensation scheme, but also by their ability to *choose* that compensation scheme, it would bias comparisons between treatment and control. If this were the case, however, we would expect a discontinuous jump in Figure 4 between piece-rate workers in the “No Offer” control group (leftmost orange dot) who have no ability to choose and piece rate workers in the lowest offer group (second-to-leftmost orange dot) who had the ability to choose but primarily opted for the piece rate. Instead, we see a gradual rise in piece-rate output, consistent with continuous selection into more generous hourly wage contracts.

¹⁷A large literature shows how “loss aversion”—a form of reference dependence—can explain the sensitivity of short-term labor supply decisions to benchmark earnings levels (Crawford and Meng, 2011; Farber, 2015; Thakral and Tô, 2021; Angrist et al., 2021; Andersen et al., 2025). While experimental offers are deliberately not framed as potential losses, workers may still feel regret if they earn less than their initial wage offer.

¹⁸Recall that the “counter” function in my experiment allows workers to observe piece-rate earnings in real time, making output highly manipulable.

One might also be concerned about framing effects biasing estimates of wage effects. Because variation between offered and effective wages is generated through unexpected top-ups, estimated wage effects could also include behavioral responses to receiving a surprise pay raise. Note, however, that most behavioral mechanisms like reciprocity through gift exchange (Falk, 2007; Akerlof, 1982) operate in the same direction as wage effects. The precise null estimates on “Accepted $\times$ Log Effective Hourly Wage” in Table 5 likely rule out these mechanisms.¹⁹

Summary of Results To summarize, experimental results provide evidence of both adverse selection and moral hazard in hourly wage contracts—workers who decline more generous wage offers have higher piece-rate productivity, workers who accept less generous wage offers have lower fixed-wage productivity, and the pooled average output of accepters and decliners falls with generosity of the wage they were offered.²⁰ In the following sections, I develop a model to investigate how these forces affect labor-market equilibrium and worker welfare. I then show how the components of this model correspond to the potential outcome marginal treatment response (MTR) functions under fixed-wage and piece-rate counterfactuals—objects that I estimate in the subsequent MTE analysis.

4 Model of Asymmetric Information in Wage Contracts

In this section, I present a model of short-term labor markets under asymmetric information. The model borrows from Einav et al. (2010a) and Herbst and Hendren (2024), who develop models of asymmetric information in health insurance markets and college financing markets, respectively.²¹

Consider a perfectly competitive labor market in which risk-neutral firms face a

¹⁹I can also rule out pure “surprise” effects from true wage effects by testing the significance of the intercept term—a response to surprises should persist for very small wage top-ups, whereas wage effects should reduce to zero as the effective wage approaches the offered wage. Adding an indicator variable for receiving *any* hourly raise (i.e., $R_i \equiv \mathbf{1}\{\text{Effective Hourly Wage} = \text{Offered Hourly Wage}\}$) in Equation (5) yields a null result.

²⁰In Appendix Table A5, I report multiplicity-adjusted p -values for tests of these three hypotheses using the multiple hypothesis testing procedure from List et al. (2019).

²¹In Appendix D, I discuss how the model can be extended to incorporate monitoring costs, partially fixed compensation, wage effects, and dynamic contracts.

fixed population of workers. Assume this population has already been pre-screened, so that workers are observably equivalent from the perspective of firms.²² Each worker, i , can produce some level of hourly output, $q_i = f(\zeta_i, e_i, \nu_i)$, which is a function of unobserved worker characteristics (ζ_i), individual effort (e_i), and random noise (ν_i). Assume firms know the data generating process, so they form unbiased beliefs about the distribution of q_i but cannot observe the ex-ante productivity of any individual worker.

Facing this population of observably identical workers with unknown productivity, firms have two options to purchase workers' labor product. One option is to buy worker output at a constant market price of p per unit, either through freelance hiring or more formal piece-rate employment.²³ Alternatively, they can offer a flat, up-front wage, w , in exchange for a claim on a worker's hourly output, q_i .²⁴

For an individual worker, i , I define the reservation wage, $\bar{w}_i$, as the minimum w at which they would accept an hourly contract over selling their labor product at the piece rate. The relative supply of hourly workers is given by

$$S(w) \equiv \Pr(\bar{w}_i \leq w). \quad (6)$$

Assuming strict monotonicity ($S(w) > S(w')$ for all $w > w'$), I index workers by a type parameter, $\theta_i \in [0, 1]$, equal to the share of the worker population willing to accept a lower wage than worker i 's reservation wage, $\theta_i \equiv S(\bar{w}_i)$. I can then rewrite a worker's reservation wage as a function their type, $\bar{w}_i = \bar{w}(\theta_i)$, where

$$\bar{w}(\theta) \equiv S^{-1}(\theta). \quad (7)$$

²²In my empirical analysis, I allow employers to set wages using public information, X_i , like individual work history and approval rating, which can act as signals of future productivity (Spence, 1973; Farber and Gibbons, 1996; Gibbons and Katz, 1991; Pallais and Sands, 2016). For simplicity, I omit these " X_i " terms from the model, which effectively considers a subpopulation of workers with observables matching a particular value, $X_i = x$.

²³Note that competitive employers cannot profitably offer piece rates that deviate from this market price, p , which is exogenously fixed. Instead, they consider buying ex-ante *claims* on labor product through fixed-wage contracts. This model is isomorphic to one in which firms sell earnings insurance at some fixed premium to self-employed workers whose labor product is competitively priced at $\$p$ per unit.

²⁴While a worker's output, q_i , can differ under hourly versus piece-rate contracts, I assume it does not vary with the level of the hourly wage, w (i.e., no wage effects). While the absence of wage effects in my empirical results would seem to validate this assumption, I include a model with wage effects in Appendix D for completeness.

To determine the profit-maximizing wage, firms consider the value of hourly output produced across worker types. I define the *marginal value* of type θ as

$$MV(\theta) \equiv E[Y_i | \theta_i = \theta], \quad (8)$$

where $Y_i = pq_i$, the incremental value of output q_i produced by worker i under an hourly contract.²⁵ $MV(\theta)$ equals the expected amount type θ 's hourly output would have earned them under the market piece rate, p . However, type θ 's reservation wages may fall below this “actuarially fair” wage (i.e., $\bar{w}(\theta) < MV(\theta)$), especially if they are risk averse.

If $\bar{w}(\theta) < MV(\theta)$, a fully informed employer could profit from offering an hourly wage of $w = \bar{w}(\theta)$ exclusively to workers of type θ . However, if employers cannot observe types, they cannot prevent workers with $\theta_i \neq \theta$ from opting into a contract offered at wage $w = \bar{w}(\theta)$. In this case, the hourly position would be accepted by all types θ_i such that $\bar{w}(\theta_i) \leq w(\theta)$. So instead of obtaining type θ 's marginal value, $MV(\theta)$, the employer would obtain their *average value*, defined as

$$AV(\theta) \equiv E[Y_i | \theta_i \leq \theta]. \quad (9)$$

The average value, $AV(\theta)$, of type θ is given by the average value of output produced among all types $\theta_i \leq \theta$. When we account for this selection into contracts, the employer's expected profits from hiring a worker at some wage w are given by

$$\Pi(w) = S(w)(AV(\theta^w) - w), \quad (10)$$

where $\theta^w \equiv S(w)$, the worker type with reservation wage equal to w .

I assume that at least one worker's marginal value exceeds their reservation wage, ($\bar{w}(\underline{\theta}) < MV(\underline{\theta})$ for some $\underline{\theta} > 0$). I further assume that $MV(\theta)$ crosses the supply curve at most once (if $\bar{w}(\bar{\theta}) > MV(\bar{\theta})$ for some $\bar{\theta}$, then $\bar{w}(\theta) > MV(\theta)$ for all $\theta > \bar{\theta}$). With these simplifying assumptions in hand, the zero-profit condition implies that

²⁵ Y_i reflects the market value of q_i units of output, or, equivalently, the amount the firm saves by not buying hourly worker i 's output at the piece rate. This measure of value is analogous to the incremental cost of insurance defined in Einav et al. (2010a). Note that any monitoring costs of observing worker output would increase this incremental value, making hourly wages more likely (see Appendix Figure A2 and Appendix D).

the equilibrium share of workers under hourly contracts, θ^{EQ} , is given by

$$\bar{w}(\theta^{EQ}) = AV(\theta^{EQ}). \quad (11)$$

In equilibrium, firms offer wage contracts up the point where the marginal worker's reservation wage, $\bar{w}(\theta^{EQ})$, is exactly equal to the average value of hourly employees' output, $AV(\theta^{EQ})$. The efficient allocation of hourly contracts, on the other hand, is given by

$$\bar{w}(\theta^{EF}) = MV(\theta^{EF}). \quad (12)$$

Under this allocation, θ^{EF} , all workers with reservation wages below their marginal values are hired for fixed-wage positions. This allocation also corresponds to the share of fixed-wage workers under a full-information equilibrium, in which firms can observe workers' potential productivity and offer them type-specific contracts.

Graphical Illustration Figure 5, Panel A illustrates the welfare impacts of adverse selection for an example population. An efficient allocation of contracts would lead to hourly employment for all types $\theta \leq \theta^{EF}$, as these workers would accept wages at or below their marginal values ($\bar{w}(\theta) \leq MV(\theta)$). However, while type θ^{EF} 's reservation wage (red line) is equal to their marginal value (blue line), an employer offering an hourly wage of $w = \bar{w}(\theta^{EF})$ would only recoup the average value (green line) among everyone accepting the offer (i.e., all $\theta \leq \theta^{EF}$). The employer could lower their wage offer, but that would drive those with the highest productivity out of the market, further reducing the contract's average value. This process continues across all types for whom $\bar{w}(\theta) > AV(\theta)$, so that the equilibrium share of workers under hourly contracts is θ^{EQ} , where $\bar{w}(\theta^{EQ}) = AV(\theta^{EQ})$. In this stylized example, roughly one-third of the population— $\theta \in (\theta_{EQ}, \theta_{EF})$ —cannot obtain hourly employment despite a willingness to work for less than their expected earnings under the market piece rate. The result is a welfare loss corresponding to the area of the region shaded in yellow, which is equal to

$$DWL = \int_{\theta^{EQ}}^{\theta^{EF}} (MV(\theta) - \bar{w}(\theta)) d\theta. \quad (13)$$

In summary, my model shows how worker selection on unobserved productivity can create a gap between the marginal and average values of labor, preventing Pareto-improving exchanges of fixed-wage contracts—workers are paid by the hour if and

only if their reservation wage is no higher than the average value of those with lower reservation wages. This information asymmetry reduces total welfare below what it would be under a full-information benchmark.

4.1 Incorporating Moral Hazard

Note that the model above allows for moral hazard effects, even if those effects are not explicitly discussed. To see how, consider worker i 's potential output values under counterfactual contracts. Specifically, let worker i 's output value under the hourly wage, currently represented as Y_i , instead be denoted by Y_{1i} . Now let Y_{0i} denote the counterfactual output value that worker i would produce if they worked under the piece rate. The moral hazard effect for worker i is given by the individual treatment effect of the hourly wage, $MH_i \equiv Y_{1i} - Y_{0i}$.²⁶ This difference in counterfactual outcomes is not explicitly shown in the model because $AV(\theta)$ and $MV(\theta)$ are defined conditional on accepting the hourly contract, and thus depend only on Y_{1i} . The profit condition (10) and welfare calculation (13) are therefore inclusive of any moral hazard effects because $MV(\theta) \equiv E[Y_{1i}|\theta_i = \theta] = E[Y_{0i} + MH_i|\theta_i = \theta]$.

While not strictly necessary to calculate welfare loss, explicitly modeling and estimating moral hazard effects is nonetheless important, especially for policy counterfactuals. As I show in Section 6, moral hazard effects are needed to assess the social value of piece-rate taxes, as the public cost of these policies must include the reduced tax revenue from potentially lower earnings among those induced into hourly wage contracts. Moreover, separately identifying moral hazard allows me to consider scenarios in which firms can mitigate the shirking response to fixed wages. For example, combining hourly wages with a partial piece-rate or commission might attenuate the disincentive effects of hourly pay, but such strategies would do little to prevent adverse selection.²⁷ To identify the model under these counterfactuals, I must explicitly separate selection from the moral hazard effects of “pure” hourly wage offers in my experiment.

To determine how market equilibrium (Equation 11) changes with and without moral hazard effects, I split Equations (8) and (9) into two pairs of curves. First, I

²⁶Strictly speaking, $Y_{1i} - Y_{0i}$ captures worker i 's overall output response to the hourly wage contract. Some of this response could result from behavioral phenomena not traditionally classified as “moral hazard.”

²⁷In Appendix D, I incorporate this mixed-compensation scenario into my framework by recasting the model as a market for supplemental hourly wages on top of a preexisting piece rate.

define marginal values of a type θ as the conditional means of potential output value with (Y_{1i}) and without (Y_{0i}) the hourly wage:

$$MV_1(\theta) \equiv E[Y_{1i}|\theta_i = \theta] \quad (14)$$

$$MV_0(\theta) \equiv E[Y_{0i}|\theta_i = \theta]. \quad (15)$$

Note that $MV_1(\theta)$ is simply a relabeling of $MV(\theta)$ from Equation (8)—it captures the expected output value under hourly wage $w = S^{-1}(\theta)$ for the worker who is indifferent between accepting or declining the offer. $MV_0(\theta)$, on the other hand, captures the expected output value of that same worker if they had instead rejected wage offer w and remained on the piece rate.²⁸ The difference between these two marginal value curves identifies the moral hazard effect for a given type:

$$MH(\theta) \equiv MV_1(\theta) - MV_0(\theta). \quad (16)$$

Similarly, the average value curve can be split into two counterfactuals:

$$AV_1(\theta) \equiv E[Y_{1i}|\theta_i \leq \theta] \quad (17)$$

$$AV_0(\theta) \equiv E[Y_{0i}|\theta_i \leq \theta]. \quad (18)$$

$AV_1(\theta)$ is equivalent to $AV(\theta)$ from Equation (9); it equals the average value of output among hourly-pay workers with lower reservation wages than type θ . $AV_0(\theta)$, on the other hand, equals the average value that would be produced by those same workers if they had instead worked under the piece rate.

Introducing dual marginal and average value curves means my model now has two counterfactual equilibria. The equilibrium condition that incorporates workers' labor-supply response to hourly pay is given by $\bar{w}(\theta_1^{EQ}) = AV_1(\theta_1^{EQ})$. By contrast, if hourly contracts have no such effects on labor supply—perhaps because employers implement increased monitoring or partial piece rates—the equilibrium allocation would instead be given by $\bar{w}(\theta_0^{EQ}) = AV_0(\theta_0^{EQ})$. Meanwhile, efficient allocations with and without moral hazard effects are given by $\bar{w}(\theta_1^{EF}) = MV_1(\theta_1^{EF})$ and $\bar{w}(\theta_0^{EF}) = MV_0(\theta_0^{EF})$, respectively. In the next section, I use MTE methods to estimate all five curves:

²⁸In a loose sense, these two curves can be thought of as bounds on output value under mixed compensation with both fixed and piece-rate components. If the piece-rate component partially mitigates moral hazard effects, the true marginal value curve would lie somewhere between $MV_0(\theta)$ and $MV_1(\theta)$.

$\bar{w}(\theta)$, $MV_1(\theta)$, $AV_1(\theta)$, $MV_0(\theta)$, and $AV_0(\theta)$. These estimates allow me not only to calculate counterfactual equilibria, but also to quantify the type-specific moral hazard effect, $MH(\theta)$, in Equation (16). As I show in Section 6, this labor-supply response is an important component of policy evaluation.

5 Estimating the Model using Marginal Treatment Effects

The model above shows how the welfare effects of asymmetric information depend on counterfactual distributions of workers’ marginal values across a range of reservation wages. In this section, I show how I can semi-parametrically identify these marginal values using a marginal treatment effects (MTE) framework.

As in Section 2, consider a potential outcomes framework in which treatment corresponds to working under an hourly contract. Let experimental wage offers, w , serve as a continuous instrument for taking up that treatment condition. Adopting the parlance of the causal inference literature, a worker’s quantile reservation wage, $\theta_i \equiv S(\bar{w}_i)$, is their “resistance to treatment” (Björklund and Moffitt, 1987; Heckman and Vytlačil, 1999, 2005, 2007). Under this framing, the marginal values defined in Equations (14) and (15) are exactly the treated and untreated marginal treatment response (MTR) functions, $MTR_1(\theta) \equiv E[Y_{1i}|\theta_i = \theta] = MV_1(\theta)$ and $MTR_0(\theta) \equiv E[Y_{0i}|\theta_i = \theta] = MV_0(\theta)$. The moral hazard effect in Equation (16) is therefore equivalent to the marginal treatment effect of the hourly contract, $MTE(\theta) \equiv E[Y_{1i} - Y_{0i}|\theta_i = \theta] = MH(\theta)$. As Figure 5, Panel B illustrates, this marginal treatment effect measures the average effect of treatment (hourly contract) among those with a given resistance to treatment (quantile reservation wage, θ_i).²⁹

The correspondence above means I can apply insights from the MTE literature to identify the model with minimal parametric assumptions. First, note that the propensity score of the wage-offer instrument is equivalent to the supply curve, $S(w)$, in Equation (6). It can be straightforwardly identified as the share of accepters across wage offers:

$$S(w) \equiv \Pr(\bar{w}_i \leq w) = \Pr(D_i = 1|W_i = w). \quad (19)$$

²⁹Kowalski (2023b) uses similar insights from the MTE literature to show how adverse selection on potential outcomes can explain differences in the estimated treatment effects of health insurance in Oregon and Massachusetts.

Next, at each propensity score, $\theta = S(w)$, average value under both hourly and piece-rate contracts can be identified from the conditional means of output value among respective accepters and decliners of the corresponding wage offer, w . So the average value curve under the hourly wage equals the average output value among those who accept the hourly wage offer that induces θ -share of workers into the hourly contract:

$$AV_1(\theta) \equiv E[Y_{1i}|\theta_i \leq \theta] = E[Y_i|S(W_i) = \theta, D_i = 1]. \quad (20)$$

Likewise, the average value curve under the piece rate equals the average output value among those who *decline* the hourly wage offer that is accepted by θ -share of workers:

$$AV_0(\theta) \equiv E[Y_{0i}|\theta_i \leq \theta] = \frac{E[Y_i|S(W_i) = 0] - E[Y_i|S(W_i) = \theta, D_i = 0](1 - \theta)}{\theta}, \quad (21)$$

where $E[Y_i|S(W_i) = 0]$ is the average output value of workers in the control group, who all work under the piece rate.³⁰ Finally, marginal values can be identified by separately differentiating take-up weighted conditional means for decliners and accepters of each offer:

$$MV_1(\theta) = \frac{\partial(E[Y_{1i}|\theta_i \leq \theta]\theta)}{\partial\theta} = \frac{\partial(E[Y_{1i}|S(W_i) = \theta, D_i = 1]\theta)}{\partial\theta} \quad (22)$$

$$MV_0(\theta) = -\frac{\partial(E[Y_{0i}|\theta_i > \theta](1 - \theta))}{\partial\theta} = -\frac{\partial(E[Y_i|S(W_i) = \theta, D_i = 0](1 - \theta))}{\partial\theta} \quad (23)$$

Intuitively, Equations (22) and (23) identify marginal values by differentiating total value, $TV(\theta) \equiv AV(\theta) * \theta$, with respect to θ under hourly and piece-rate counterfactuals. To estimate these objects, I apply the local polynomial regression approach from Carneiro et al. (2011). Estimation details are provided in Appendix E.

5.1 Results

Figure 6 plots semi-parametric estimates of supply and value curves under both hourly wage and piece-rate counterfactuals. On the horizontal axis, the type parameter, θ , corresponds to quantiles of workers' hourly reservation wages. The red line plots hourly reservation wage at each quantile, $\bar{w}(\theta)$, which equals the inverse of the labor

³⁰By offering no alternative to the piece rate, the control condition effectively allows for “identification at infinity”—the average piece-rate output among workers of all reservation wages (Heckman, 1990; Chamberlain, 1986).

supply curve implied by first-stage logit estimates from Table 3, $\bar{w}(\theta) \equiv S^{-1}(\theta)$.³¹ In Panel A, the green and blue lines plot the average and marginal value curves under hourly wages, $AV_1(\theta) \equiv E[Y_{1i}|\theta_i \leq \theta]$ and $MV_1(\theta) \equiv E[Y_{1i}|\theta_i = \theta]$, respectively. In Panel B, green and blue lines plot the average and marginal value curves under the piece rate, $AV_0(\theta) \equiv E[Y_{0i}|\theta_i \leq \theta]$ and $MV_0(\theta) \equiv E[Y_{0i}|\theta_i = \theta]$, respectively.³² In both panels, value curves slope upward, reflecting the negative selection into fixed wages observed in Section 3. Indeed, the statistically significant coefficients on “Accepted $\times$ Log Hourly Wage Offer” and the positive, marginally significant coefficient on “Declined $\times$ Log Hourly Wage Offer” in Table 5 provide reduced-form evidence against flat value curves, especially under hourly wages.

Full Information Benchmark In both panels of Figure 6, the majority of workers produce labor at marginal values, $MV(\theta)$, that exceed their hourly reservation wages, $\bar{w}(\theta)$. In Panel A, 56 percent (SE=1.89) of workers have $\bar{w}(\theta) < MV_1(\theta)$. This share represents the equilibrium that would exist if employers were fully informed of workers’ potential output and set fixed wages to incorporate their moral hazard response. In the absence of moral hazard effects (Panel B), the share of workers with $\bar{w}(\theta) < MV_0(\theta)$ is 58 percent (SE=1.88), simulating an environment in which firms can eliminate shirking through partial piece rates or other means (see Section 4.1).

The apparent willingness of workers to accept fixed wages ($\bar{w}(\theta)$) below their actuarially fair, breakeven values ($MV(\theta)$) is consistent with existing evidence of ostensibly risk-averse behavior in low-stakes settings (Falk et al., 2018; Andersen et al., 2008; Harrison et al., 2007). Under standard utility functions, such behavior implies implausibly high levels of risk aversion in higher-stakes environments (Rabin, 2000), leaving researchers to rely on behavioral explanations like myopia (Kahneman and Tversky, 1979; Prelec, 1998), mental accounting (Rabin and Weizsäcker, 2009; Barberis et al., 2006), and reference dependence (Kahneman and Tversky, 1991).³³ In my setting, systematically biased beliefs provide yet another potential explanation for seemingly risk-averse decisions—a risk-neutral worker might prefer $w < MV(\theta)$ if they

³¹Appendix Figure A3 reports analogous estimates from an alternative specification that replaces the logit first stage with a monotone-spline first stage in the offered wage. Appendix E provides estimation details.

³²Note that error bands in each plot reflect mechanically correlated sampling error between estimated value curves.

³³In my experiment, partial tests for reference dependence or loss aversion yield null results. See Section 3 and Appendix Figure A1.

undervalue their productive potential. Importantly, my design allows me to remain agnostic about which of these forces is driving workers’ preferences for fixed wages. Because semi-parametric estimates of marginal and average value curves rely entirely on revealed preferences over contracts, I can determine both competitive equilibrium and its full-information counterfactual without imposing explicit structure on worker utility or the distribution of parameters governing it. Workers’ choices could be driven by any combination of risk aversion, myopia, or other behavioral phenomena, and the market outcomes of those choices would remain the same.³⁴

Partial Unraveling from Adverse Selection Despite most workers’ willingness to sacrifice expected earnings in exchange for a fixed wage contract, the most productive of these workers remain on piece rates due to adverse selection. This inefficiency can be seen in the divergence between the marginal and average value curves in Figure 6. In an equilibrium with both adverse selection and moral hazard (Panel A), only 49 percent (SE=1.56) of workers can find hourly positions, compared to a full-information benchmark of 56 percent (SE=1.89). The resulting welfare loss from this attenuation in hourly work is an estimated \$0.05 (SE=0.017) per hour of labor, represented by the shaded orange region in Figure 6A.³⁵ Using a back-of-the-envelope estimate of roughly 35 billion annual paid worker-hours in online clerical and data-entry gig work, this per-hour loss implies aggregate annual welfare costs of about \$1.7 billion globally.³⁶

³⁴Note that, even if they have no impact on equilibria, non-classical explanations of worker choice may have different welfare implications. Under many behavioral theories these implications depend upon which choices reflect “true” preferences and which reflect “mistakes” (Goldin and Reck, 2022). Rather than weigh in on this question, this paper takes a classical approach to welfare and policy analysis, assuming that a worker’s willingness to accept wage offers below their breakeven value corresponds to a revealed preference for fixed wages rather than systematic errors.

³⁵Because standard errors are jointly computed over bootstrap replications, they account for mechanically correlated sampling error between $MV(\theta)$ and $AV(\theta)$ curves. Note that estimates of equilibrium, efficient allocations, and welfare loss are not affected by extrapolated portions of estimated value curves, which are denoted by dashed lines in Figure 6.

³⁶I obtain the 35 billion annual-hours estimate by combining the lower end of the World Bank’s estimate of 154 to 435 million global online gig workers with the World Bank’s estimate that clerical and data-entry tasks account for 23 percent of online labor demand, and the ILO’s estimate that online platform workers spend an average of 27 hours per week on platform work, of which roughly 8 hours are unpaid (World Bank, 2024; International Labour Organization, 2021). This yields $154 \times 0.23 \times (27 - 8) \times 52 \approx 35$ billion annual paid worker-hours. For comparison, the broader U.S. category of office and administrative support occupations accounted for 18.5 million jobs, or 12.2 percent of total employment, in 2024; multiplying by 1,900 to 2,080 hours per job implies roughly 35 to 38 billion annual paid hours in the United States alone (U.S. Bureau of Labor Statistics, 2024).

In the absence of moral hazard effects (Panel B), the equilibrium share of hourly work under adverse selection is 51 percent (SE=2.37), compared to a full-information benchmark of 58 percent (SE=1.88). Estimated welfare loss under this no-shirking counterfactual is \$0.04 (SE=0.038)—slightly smaller and less precise than estimates inclusive of moral hazard responses.

Selection on Moral Hazard The two sets of equilibria shown in Figure 6 are driven by differences in selection on hourly-wage marginal values, $MV_1(\theta)$, and selection on piece-rate marginal values, $MV_0(\theta)$. Figure 7 plots these differences across the distribution of worker types, $MV_1(\theta) - MV_0(\theta)$, which is equivalent to the marginal treatment effect of hourly wages. In the context of the model, this curve represents how the (marginal) moral hazard effect of an hourly wage contract changes with workers’ quantile reservation wage. Its shape suggests that selection on moral hazard is non-linear: Among workers with below-median reservation wages, those with marginally stronger preferences for fixed wages (lower θ) have a slightly higher propensity to shirk (more negative treatment effect), consistent with patterns of selection in insurance markets (Einav et al., 2013). Among those with above-median reservation wages, however, shirkers tend to prefer piece rates—those most resistant to hourly wages are most likely to shirk once they have it. This pattern could be explained by heterogeneous risk preferences: If tolerance for earnings risk correlates with tolerance for reputational risk from excessively low output, those who prefer the variability of piece-rate pay may also be more likely to shirk under fixed wages because they are less fearful of poor performance reviews.

Regardless of the underlying mechanism, moral hazard estimates play an important role in policy evaluation. Piece-rate taxes might mitigate adverse selection into fixed wages, but such policies must balance the welfare benefits they provide to the marginal fixed-wage worker with the distortionary costs of encouraging that worker to shirk. I explore this tradeoff in the following section.

A mechanical extrapolation of that employment share to the 3.51 billion employed persons reported worldwide in 2024 implies on the order of 0.8 trillion annual paid hours globally (International Labour Organization, 2024).

6 Policy Implications: MCPF of Piece-Rate Taxes

Section 5 showed that distortionary costs of adverse selection in hourly work result in a welfare loss of approximately \$0.05 (SE=0.017) per hour of labor among data-entry workers, translating to an aggregate loss of roughly \$1.7 billion or more per year. However, evaluating policies designed to address this market inefficiency depends less on the size of the welfare loss and more on how efficiently it can be recovered. When targeted at the right populations, policies aimed at small market inefficiencies can sometimes generate a large social return to public expenditure. In this section, I measure this social return on piece-rate taxes in data-entry using the marginal value of public funds (MVPF), defined as the welfare benefit per dollar of government expenditure (Hendren and Sprung-Keyser, 2020).³⁷ Note that this MVPF is equivalent to the marginal *cost* in social welfare per dollar of revenue *raised* through such taxes.³⁸ For this reason, I refer to this quantity as the marginal *cost* of public funds (MCPF) associated with the tax.

6.1 MCPF Derivation

Consider an ad-valorem piece-rate tax, ρ , assessed on the value of labor product produced under the piece rate. The MCPF of this tax is given by

$$MCPF_{\text{Tax}}(\rho) \equiv \frac{WTP(\rho)}{NR(\rho)}, \quad (24)$$

where $WTP(\rho)$ is the aggregate private willingness-to-pay to *avoid* the tax, and $NR(\rho)$ is net government revenue raised by the tax. Importantly, $NR(\rho)$ includes both the tax's direct transfer to the government and any fiscal externalities it imposes on long-term revenue.

To calculate welfare costs, $WTP(\rho)$, and net revenue, $NR(\rho)$, I parameterize the policy as a wedge on piece-rate transactions: For each dollar of labor product compensated under the piece rate, the government collects an additional ρ dollars in tax revenue. Under this wedge, procuring labor product through the piece rate

³⁷In Appendix F, I calculate MVPFs for hourly wage subsidies as an alternative to piece rate taxes. Results are reported in Appendix Figure A5.

³⁸For revenue-raising policies, a lower-valued MVPF implies that piece-rate taxes are more efficient source of public funds, with values below one implying they are more socially efficient than a non-distortionary tax (Boning et al., 2024).

becomes $1 + \rho$ times as expensive as in the baseline, so the equilibrium condition under tax ρ is given by $\bar{w}(\theta^\rho) = (1 + \rho)AV_1(\theta^\rho)$, where θ^ρ is the equilibrium share of hourly workers under the tax.³⁹ The direct fiscal consequence of the tax would therefore be to increase government revenue by $\rho MV_0(\theta)$ for all types $\theta \in [\theta^\rho, 1]$, reflecting total tax receipts from piece-rate production in the tax equilibrium. However, this increased revenue is partially offset by the indirect costs of inducing more workers into hourly pay—a tax on piece rates could lead to lower earnings and decreased tax revenue. Net government revenue, $NR(\rho)$, from the piece-rate tax is therefore given by

$$NR(\rho) = \underbrace{\int_{\theta^\rho}^1 \rho MV_0(\theta) d\theta}_{\text{Direct Revenue from Transfer}} + \underbrace{\int_{\theta^{EQ}}^{\theta^\rho} \tau MH(\theta) d\theta}_{\text{Fiscal Externality from Moral Hazard}}, \quad (25)$$

where τ represents the baseline tax rate on earned income.⁴⁰

The social cost of raising revenue $NR(\rho)$ is given by the aggregate private willingness-to-pay to avoid the tax. This willingness-to-pay, $WTP(\rho)$, has two components: First, piece-rate production with $\theta \in [\theta^\rho, 1]$ would avoid direct tax payments of $\rho MV_0(\theta)$ in the absence of the tax. Second, workers with $\theta \in [\theta^{EQ}, \theta^\rho]$ would lose $MV_1(\theta) - \bar{w}(\theta)$ of insurance value from hourly positions supported by the tax. The total welfare cost of piece-rate tax, ρ , is therefore given by

$$WTP(\rho) = \underbrace{\int_{\theta^\rho}^1 \rho MV_0(\theta) d\theta}_{\text{Direct Tax Savings}} - \underbrace{\int_{\theta^{EQ}}^{\theta^\rho} (MV_1(\theta) - \bar{w}(\theta)) d\theta}_{\text{Lost Insurance Value}}. \quad (26)$$

With Equations (25) and (26) in hand, I can write $MCPF_{\text{Tax}}(\rho)$ —the MCPF of piece-rate tax ρ —as

$$MCPF_{\text{Tax}}(\rho) \equiv \frac{WTP(\rho)}{NR(\rho)} = \frac{\int_{\theta^\rho}^1 \rho MV_0(\theta) d\theta - \int_{\theta^{EQ}}^{\theta^\rho} (MV_1(\theta) - \bar{w}(\theta)) d\theta}{\int_{\theta^\rho}^1 \rho MV_0(\theta) d\theta + \int_{\theta^{EQ}}^{\theta^\rho} \tau MH(\theta) d\theta}. \quad (27)$$

³⁹Recall that the marginal and average values of hourly workers' labor product are defined as the amount the firm saves by not procuring that labor product directly through the piece rate (see Footnote 25). Under this tax wedge, each dollar of piece-rate labor product generates an additional ρ dollars of tax liability, so the corresponding hourly value curves scale by a factor of $1 + \rho$. Note that this equilibrium condition assumes the price of labor product, p , is exogenously fixed. In Appendix F and Appendix Figure A4, I discuss the second-order effects that piece-rate taxes might have through changes to p in general equilibrium.

⁴⁰Baseline tax rates are calibrated to $\tau = 0.2$.

Equation (27) reveals the trade-off faced by policymakers promoting hourly wage contracts. The marginal social benefit of an additional hourly contract depends on the relative magnitudes of its insurance value to the marginal worker and that worker’s propensity to shirk. More generally, this trade-off highlights the importance of separating adverse selection from moral hazard in markets with asymmetric information—misattributing one for the other can lead to suboptimal policy decisions.

The green line in Figure 8, plots estimates of $MCPF_{\text{Tax}}(\rho)$. Estimated MCPFs increase with the size of the tax because the first worker induced into hourly pay has the highest risk premium among non-hourly workers. The vertical line denotes the tax that achieves the hourly supply share found in a full-information equilibrium. This “efficient” tax rate is equal to 20 percent (SE=3.52) and results in an MCPF equal to 0.94 (SE=0.024).

6.2 Calculating Optimal Piece-Rate Tax

While the above analysis helps identify the range of potential MCPFs associated with piece rate taxes, it does not solve for the welfare-maximizing tax rate, ρ^* . In general, comparisons of MVPFs between mutually-exclusive policies that endogenously differ in scale can lead to suboptimal policy choices. In this instance, the lowest-MCPF tax would generate a vanishingly small amount of total revenue because the first person induced into the hourly wage has the highest possible welfare gain from leaving the piece rate.

To determine the optimal tax, I use Equations (25) and (26) to find the value, ρ^* , that minimizes aggregate net welfare costs:

$$\rho^* \equiv \underset{\rho}{\operatorname{argmin}} \{WTP(\rho) - \eta NR(\rho)\}. \quad (28)$$

where η represents the marginal value of government revenue—the highest MVPF among policies to which tax revenue might be allocated. The first order conditions for (28) imply

$$MCPF_{d\text{Tax}}(\rho^*) \equiv \frac{WTP'(\rho^*)}{NR'(\rho^*)} = \eta. \quad (29)$$

$MCPF_{d\text{Tax}}(\rho)$ is the MCPF for a *marginal increase* in piece-rate tax.⁴¹ Equation (29) provides a prescription for achieving the optimal tax rate—the one that minimizes net

⁴¹Appendix F provides details on the derivation of $MCPF_{d\text{Tax}}(\rho)$.

welfare costs. Policymakers should increase the tax until the MCPF of a marginally higher tax equals the marginal value of a one-dollar increase in government revenue—the MVPF of a tax reduction or policy to which the revenue might be directed.

The pink line in Figure 8 plots estimates of $MCPF_{dTax}(\rho)$. The MCPF of marginally higher taxes increases with the tax rate, reaching one at a piece-rate tax of 19 percent (SE=4.43). If we allow for a marginal value of government funds greater than one ($\eta > 1$), the optimal tax would increase from 19 to the value of ρ at which $MCPF_{dTax}(\rho) = \eta$.

In conclusion, MCPFs for piece-rate taxes compare favorably with many alternative revenue sources. By mitigating adverse selection costs, taxing output-based pay at a rate of 19 percent or less can raise government revenue at least as efficiently as a distortionless tax. At these levels, each dollar of piece-rate tax revenue carries a net social cost as low as \$0.87 and no higher than \$1.00. While these point estimates are specific to the data-entry workers in my setting, they may nonetheless inform priors on the welfare impact of taxing tips, commissions, or bonuses in other labor markets. For example, my findings suggest that recent legislation eliminating taxes on tips is likely to amplify the welfare costs of adverse selection into fixed wages.⁴² Indeed, the MCPFs calculated in this section can be thought of as quantifying the inefficiency costs one would expect from OBBB-like tax exemptions on piece rates paid to data labeling workers.

7 Implications for Other Labor Markets

Any job with an uncertain, effort-intensive labor product is susceptible to the forces of moral hazard and adverse selection. Perhaps as a consequence, labor markets most vulnerable to these forces are often dominated by self-employment, freelance work, or piece-rate compensation.⁴³ But while this paper sheds light on the potential

⁴²The One Big Beautiful Bill (OBBB) Act, signed into law July 4, 2025 grants tax-advantaged status to tips over fixed-wage compensation in a variety of industries (119th United States Congress, 2025).

⁴³Failed attempts at “no-tipping restaurants” provide a potential case study: After replacing tipped earnings with fixed wages, many restaurants struggled to maintain profitability and ultimately reverted course, often citing an exodus of qualified servers as the cause (Kadvany, 2022b; Moskin, 2020; Dunn, 2018). One San Francisco restaurant noted that “[Server positions] have been harder to fill. Many veteran servers weren’t interested, saying they could make double elsewhere with tips...As a result, many of the people who work in the dining room started with little to no restaurant experience” (Kadvany, 2022a).

distortionary effects of asymmetric information in these settings, my estimates are not intended to measure welfare losses beyond data-entry or clerical gig work. Such limits to generalizability are ubiquitous in applied research on worker incentives—whether they come from agricultural workers (Brune et al., 2022; Bandiera et al., 2010), call centers (Mas and Pallais, 2017; Nagin et al., 2002), cashiers (Mas and Moretti, 2009), or automotive glass repairers (Lazear, 2000), specific estimates of parameters concerning worker productivity are usually difficult to generalize beyond narrowly defined labor markets.⁴⁴ While my experiment is not immune to these limitations, it has several features that help maximize external validity. As discussed in Section 2.4, the experimental job posting was designed to mimic a private firm seeking “AI Taskers,” and the typing task requires a dimension of effort and skill that is frequently requested by AI-based businesses. Moreover, participants were not informed that they were part of an experiment until after they performed the task.

In Appendix G, I further assess the external validity of my design using the SANS conditions from List (2020). This assessment establishes how, at a minimum, my results speak to information asymmetries in the large and growing online labor markets for data-labeling services. I also show how my estimates can be interpreted as approximate lower bounds on welfare loss in many other labor markets, especially those for short-term gig work. Nonetheless, policies aimed at addressing information asymmetries outside of data-entry work warrant further evidence targeted to those settings. Luckily, this paper provides a flexible theoretical and empirical framework for gathering this evidence. The following calibration exercise provides an example application.

7.1 Application to Rideshare Drivers: Calibration from Angrist et al. (2021)

To demonstrate how my framework might be applied to other settings, I calibrate my model to approximate welfare loss from adverse selection in the market for rideshare drivers. My calibration exercise uses estimates from Angrist et al. (2021), who conduct an experiment that offers Uber drivers the opportunity to increase their commission on ride fares in exchange for making fixed weekly payments that do not vary with

⁴⁴For example, Herbst and Mas (2015) find that estimates of peer effects on worker output vary dramatically between study settings.

earnings. Offering this “taxi-style” compensation structure effectively gives the driver an opportunity to buy back a share of the fare revenue they generate. Viewed through this lens, opting into a taxi-style lease is the same as opting *out* of a (partial) fixed wage contract—a driver who accepts a lease offer gives up some fixed weekly income in favor of a higher share of their labor product. This equivalence means I can calibrate my model using estimates of selection *out of* taxi leases from Angrist et al. (2021) to approximate welfare losses from inefficient compensation in the market for rideshare drivers.⁴⁵

Appendix Figure A6 plots results from this exercise. The horizontal axis denotes the share of drivers declining weekly lease offers. The red curve plots the inverse supply of lease-free drivers, $\bar{w}_B(\theta) \equiv S^{-1}(\theta)$. The blue line plots the marginal value curve, $MV(\theta) \equiv E[Y_i|\theta_i = \theta]$, where Y_i denotes driver i ’s weekly revenue from ride fares. The green line plots the average value curve, $AV(\theta) \equiv E[Y_i|\theta_i \leq \theta]$. The equilibrium share of lease-free contracts is $\theta^{EQ} = 54.7$ percent, while the efficient share of lease-free contracts we would expect in a full-information equilibrium is $\theta^{EF} = 59.4$ percent. The difference of $\theta^{EF} - \theta^{EQ} = 4.7$ percentage points represents potential drivers who would generate profitable levels of fare revenue if they were charged a lower proportional fee or paid some additional fixed weekly compensation that placed a higher value on their labor product. However, the cost of adverse selection among less productive drivers would make offering this compensation unprofitable. The corresponding welfare loss, shaded in yellow, is \$1.32 per week, or approximately \$0.09 per hour worked, comparable to my estimates of welfare loss in the market for online data-entry. Appendix Figure A7 plots MCPFs for taxes on driver fares, which range between .86 and .95—better than a distortionless tax.

There are several limitations to this calibration exercise. Calibrated curves, particularly the average value curve, rely on transformations across a small number of point estimates. Linear extrapolating market-wide selection patterns from these points comes with strong functional-form assumptions, and the absence of standard errors prevents any meaningful discussion of statistical significance. Moreover, as I discuss in Section 1, a precise accounting for welfare costs of selection must consider the value of labor product “off the equilibrium path,” but the experimental sample in Angrist et al. (2021) consists of *existing* Uber drivers. In this sample, observed selection into taxi-style lease offers likely excludes drivers who desire it most—the ones who

⁴⁵Calibration details are provided in Appendix H.

are already working as taxi drivers! More generally, because the welfare consequences of asymmetric information require market-wide estimates of treatment and selection, studies of workers under existing employment contracts are likely to understate the effects of information asymmetries. Nonetheless, this calibration exercise provides an example of how my framework can be applied to a variety of labor markets where workers selection on potential productivity can prevent efficient contracting.

8 Conclusion

This paper uses an experimental approach to estimate the equilibrium and welfare effects of moral hazard and adverse selection in fixed-wage contracts. My experiment offers workers a choice between a piece rate and a randomized hourly wage, allowing me to separately identify selection and treatment effects of wage contracts. Using experimental wage offers as an instrument for hourly wage take-up, I find evidence of both moral hazard and adverse selection. Hourly wage contracts reduce worker productivity by an estimated 8.74 percent relative to the mean. Meanwhile, a 10 percent increase in the hourly wage offer attracts a marginal worker whose productivity is higher by 1.90 percent of mean worker output.

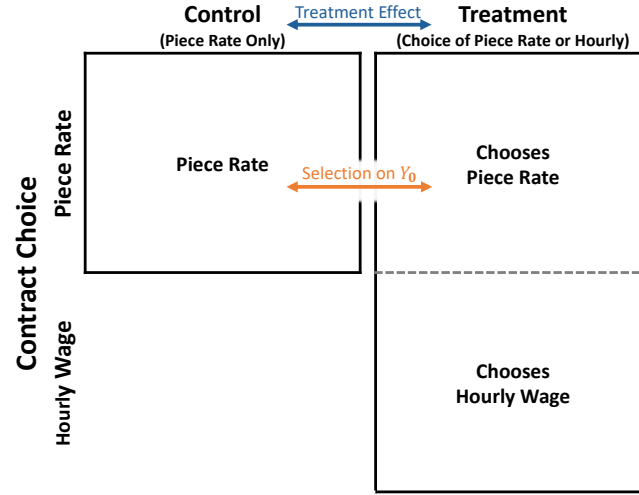
I place these experimental estimates into a theoretical framework that shows how the provision of hourly employment contracts is determined by two factors: a worker's reservation wage for fixed compensation and the average output of workers with comparatively lower reservation wages. I semi-parametrically identify these objects using a marginal treatment effects (MTE) framework. My estimates imply that information asymmetries lead to an underprovision of fixed-wage contracts, resulting in a welfare loss of \$0.05 per hour worked. To investigate the policy implications of this welfare loss, I calculate the marginal costs of public funds (MCPFs) for taxes on performance-based pay. My estimates suggest that a 19-percent tax on performance-based pay can efficiently raise government revenue by correcting the market inefficiencies associated with adverse selection.

My findings point to several directions for future research. As I discuss in Section 7, my framework can be applied to a variety of other labor markets, especially those characterized by self-employment, freelance work, or piece-rate compensation. It could also be applied to alternative contracts in the broader labor market. For example, (Ganong et al., 2025) document large month-to-month volatility in earnings

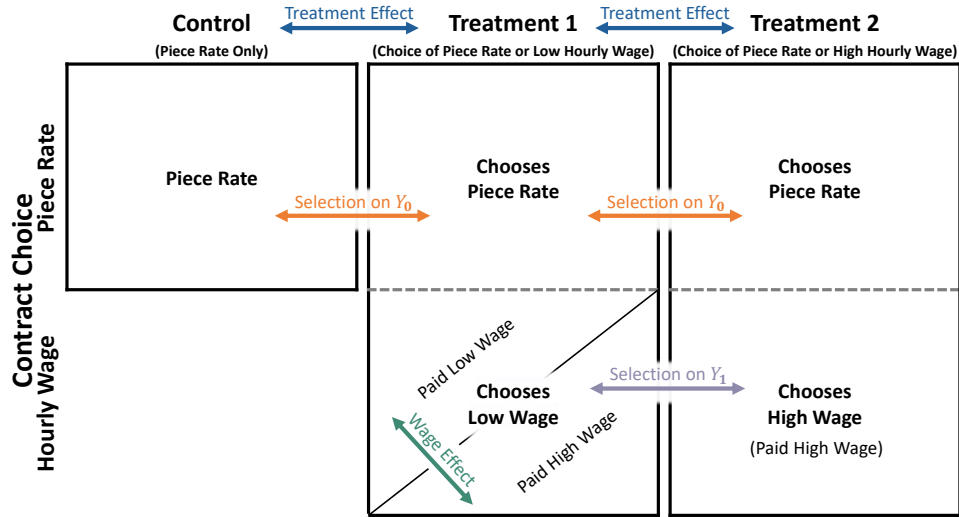
among stably employed hourly workers, apparently driven by fluctuations in firms' demand for hours. My methods could help determine why employers neglect to insure these workers through monthly or annual contracts. Future work could also extend my tax analysis to other labor-market policies under asymmetric information. A binding minimum wage, for instance, may act as an “insurance mandate” that pools workers with different latent productivities and mitigates adverse selection. Portable benefits programs and employment classification rules may create similar opportunities. Designing policies to better address information asymmetry problems in wage contracts could meaningfully improve the lives of millions of workers.

Figures and Tables

Figure 1: Experimental Design



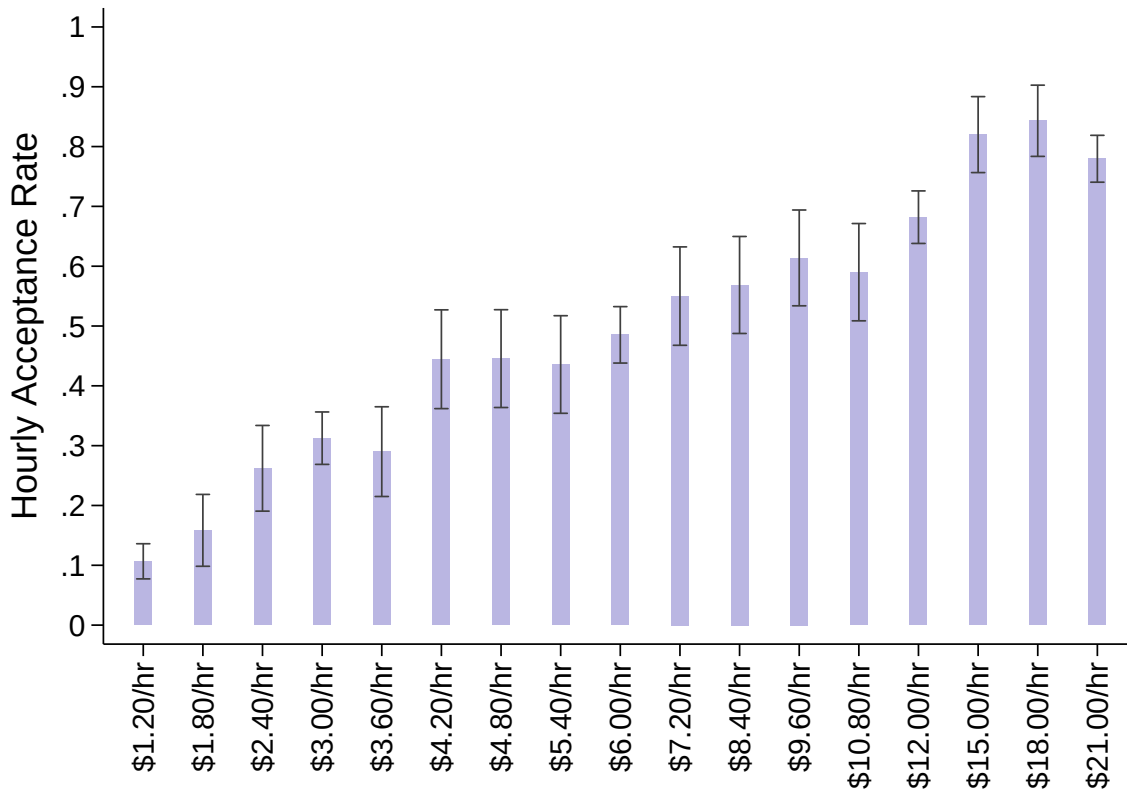
(A) Single Treatment Example



(B) Multiple Treatment Example

Note: This figure provides a graphical illustration of my experimental design, simplified to include only two (Panel A) or three (Panel B) wage-offer conditions. In each panel, columns denote experimental groups with different offer conditions. The control group is only compensated through the piece-rate contract, and treatment groups are offered a choice between the piece-rate contract and different hourly wages. Treatment groups are separated into those who accept the piece-rate contract (upper boxes) and those who accept their respective hourly wage offers (lower boxes). The diagonal split in the bottom box of Treatment 1, Panel B represents the second stage of randomization, in which some workers accepting the low hourly wage are promised the higher wage before they begin the task. Arrows denote comparisons that, after scaling by take-up, identify treatment effects (blue), selection on output under the piece rate, Y_0 (orange), selection on output under hourly wages, Y_1 (purple), and wage effects (teal). See Section 2 for details.

Figure 2: Hourly Wage Take-Up



Note: This figure reports hourly-wage acceptance rates by treatment group. The y-axis measures the share of workers in each group who declined the \$0.03 piece rate in favor of the hourly wage offer displayed on the x-axis. Note that both piece rate and hourly wage options are offered as supplements to a base wage of \$12.00 per hour. Bands indicate 90% confidence intervals.

Figure 3: Worker Output Value by Treatment Offer and Acceptance Status



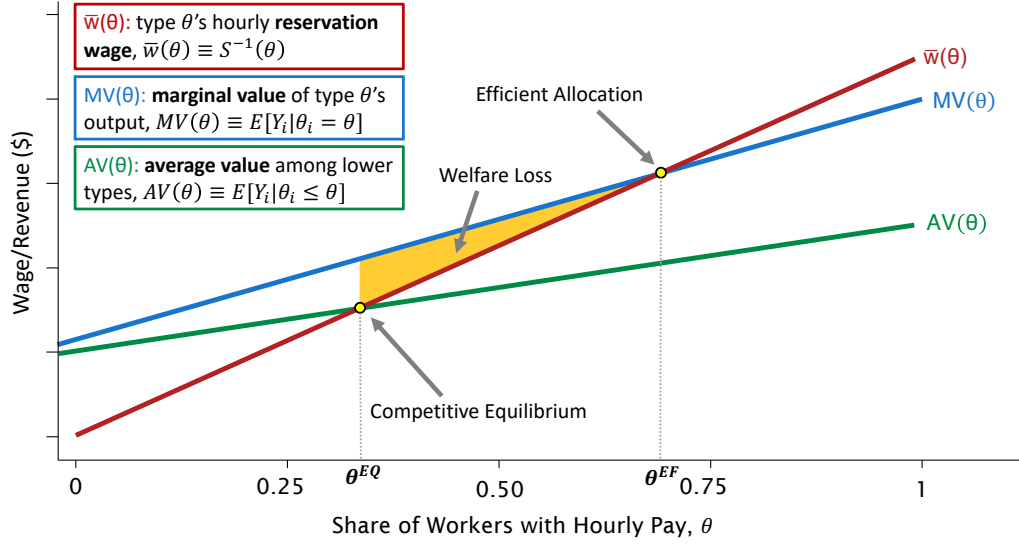
Note: This figure shows mean worker output value by wage-offer groups and compensation choice. “Output value” is defined as the number of typed sentences per hour multiplied by \$0.03. Control and treatment groups are labeled on the x-axis. Blue circles measure mean output values among all individuals in each group. Orange bars measure mean output values among those who were paid the \$0.03 piece rate. Dark green bars measure mean output values among those who chose the hourly wage offer and received a randomized top-up above the offered rate, bringing their hourly wages to the \$21.00 per hour maximum. Light green bars measure mean output values among those who chose the hourly wage offer and did not receive a top-up. Note that both piece rate and hourly wage options are offered as supplements to a base wage of \$12.00 per hour. Bands indicate 90% confidence intervals.

Figure 4: OLS Estimates of Selection on Output Value by Wage Offer

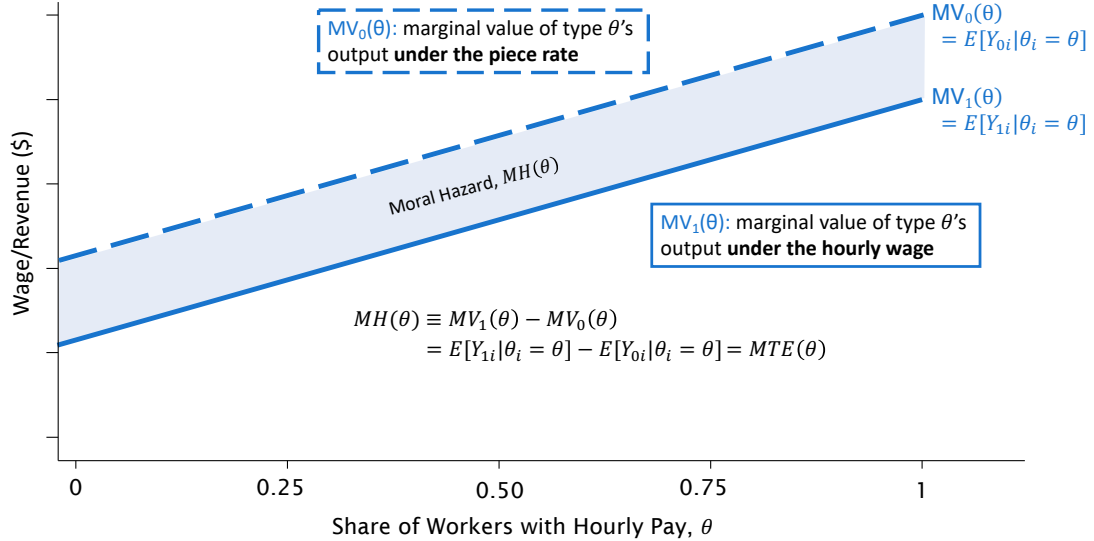


Note: This figure plots coefficients from an OLS regression of output value against the full set of dummy variables for each experimental wage offer, controlling for log effective wages among hourly workers (inclusive of top-ups) as well as task timing. Orange dots represent coefficients on hourly wage offers interacted with an indicator for remaining on the piece rate. Green diamonds represent coefficients on hourly wage offers interacted with an indicator for accepting the offer. Lines represent 90% confidence intervals.

Figure 5: Model of Asymmetric Information in Wage Contracts



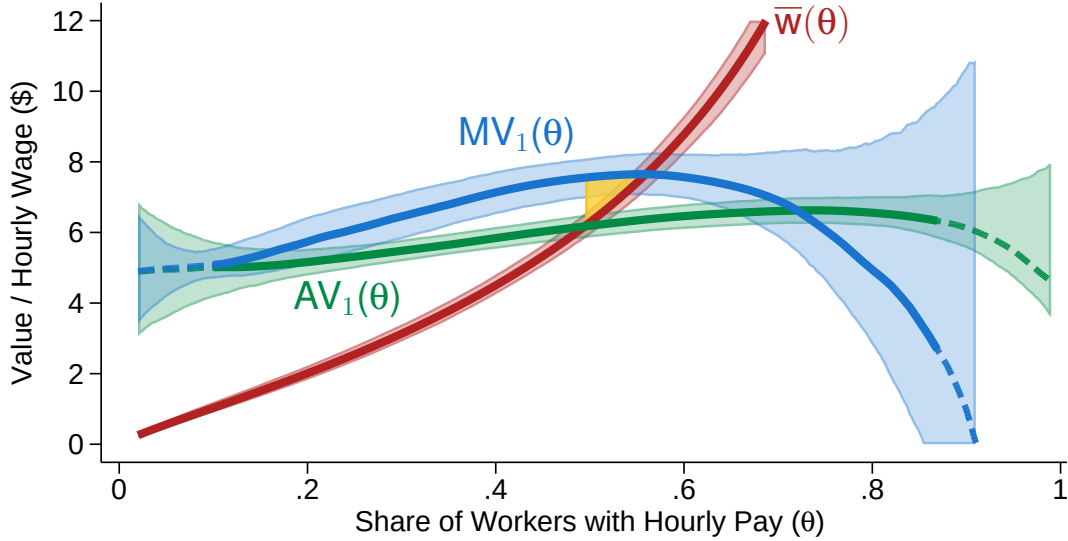
(A) Welfare Loss from Adverse Selection



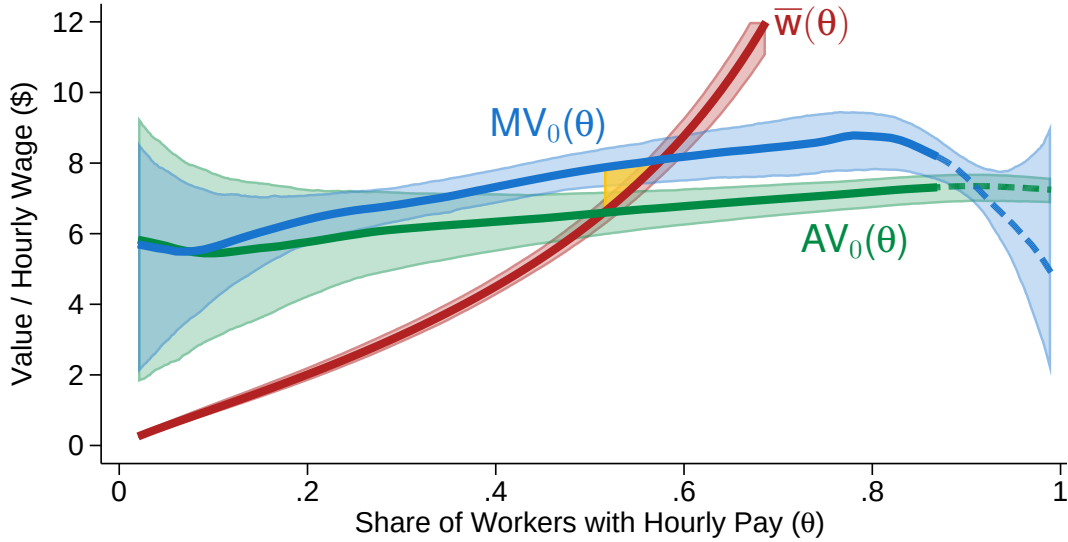
(B) Incorporating Moral Hazard

Note: This figure provides a graphical example of my model. Panel A illustrates welfare loss from adverse selection. On the horizontal axis, types, θ , are enumerated in ascending order based on their hourly reservation wage, $\bar{w}_i$. The blue line plots the marginal value curve, $MV(\theta) \equiv E[Y_i|\theta_i = \theta]$, where Y_i denotes output value under the hourly wage. The red line plots hourly reservation wage, $\bar{w}(\theta) \equiv S^{-1}(\theta)$. The green line plots the average value curve, $AV(\theta) \equiv E[Y_i|\theta_i \leq \theta]$. The orange region corresponds to the welfare loss from adverse selection into hourly wages. Panel B incorporates moral hazard into the model. The solid and dashed blue lines denote marginal values of potential output under hourly wages, $MV_1(\theta) \equiv E[Y_{1i}|\theta_i = \theta]$, and under piece rate, $MV_0(\theta) \equiv E[Y_{0i}|\theta_i = \theta]$, respectively. The difference between the two marginal value curves identifies the moral hazard effect for a given type, $MH(\theta) \equiv MV_1(\theta) - MV_0(\theta)$, which is equivalent to the marginal treatment effect of the hourly contract among those whose resistance to treatment (quantile reservation wage, $\theta_i \equiv S(\bar{w}_i)$) is equal to the propensity score (share of hourly workers, $\theta = S(w)$) for their assigned instrument (wage offer, W_i).

Figure 6: Estimates of Marginal and Average Value Curves



(A) Potential Value Under Hourly Wage



(B) Potential Value Under Piece Rate

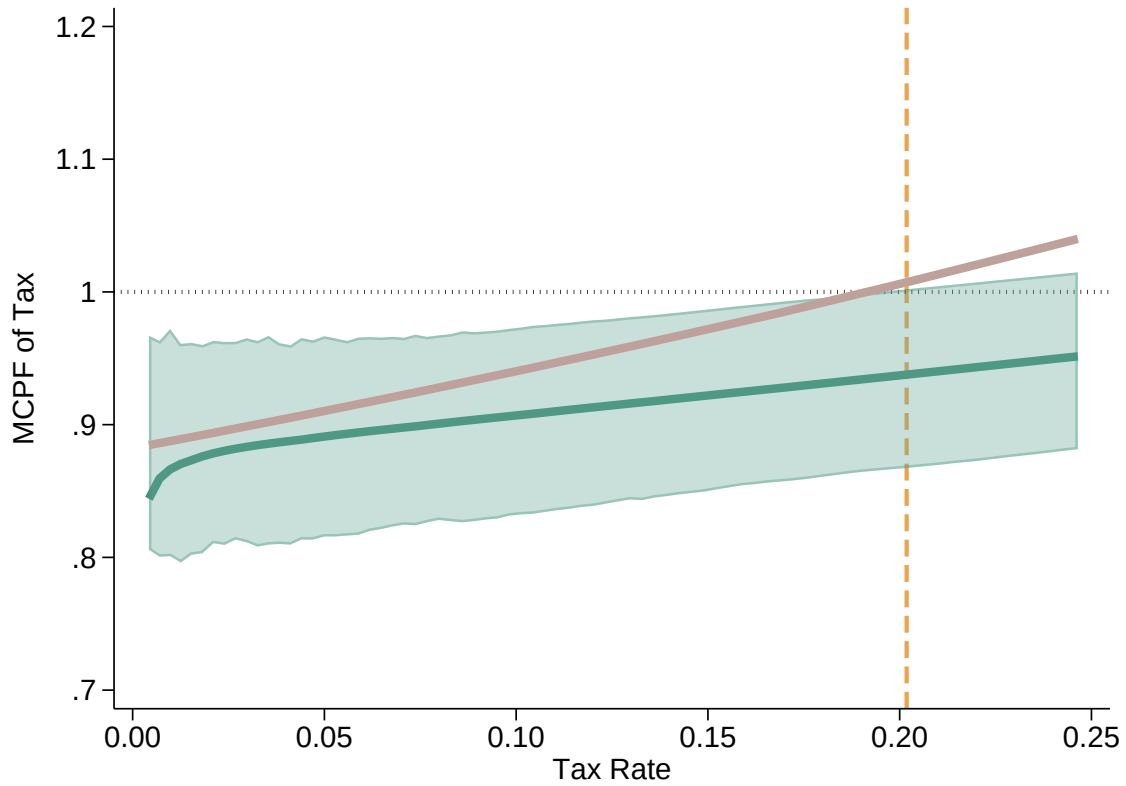
Note: This figure plots estimated supply and value curves, where output values reflect the number of correctly typed sentences multiplied by the piece rate. In the top panel, the blue and green lines plot semi-parametric estimates of the marginal value, $MV_1(\theta)$, and average value $AV_1(\theta)$, under hourly wages, as defined in Figure 5. In the bottom panel, blue and green lines plot these same curves ($MV_0(\theta)$ and $AV_0(\theta)$) under a piece-rate counterfactual. In both panels, the red line plots estimated hourly supply from a logit regression of hourly take-up against log experimental wage offers. Areas in yellow denote estimated welfare loss from adverse selection into hourly wages, estimated to be \$0.05 (SE=0.017) and \$0.04 (SE=0.038) per hour of labor, respectively. Value curves are estimated using a second-degree local polynomial regression (bandwidth=0.2) of residualized hourly output value against predicted hourly supply. Dashed portions of each line represent regions outside the support of observed propensity scores over which local polynomials were extrapolated. Shaded regions represent 90% confidence intervals, constructed via bootstrap resampling. Note that standard errors on welfare estimates account for mechanically correlated sampling error between $MV(\theta)$ and $AV(\theta)$ curves.

Figure 7: Estimates of Marginal Treatment Effects (Moral Hazard)



Note: This figure plots estimated marginal treatment effects of hourly wages on worker output value. Estimates are obtained using local polynomial regressions of worker output value against propensity score (i.e. hourly supply share), as described in Section 5. Solid lines denote $MH(\theta) \equiv MV_1(\theta) - MV_0(\theta)$ —the difference in the marginal worker’s potential output value under an hourly wage versus the piece rate. Value curves are estimated using a second-degree local polynomial regression (bandwidth=0.2) of residualized hourly output value against predicted hourly supply. Dashed portions of the line represent regions outside the support of observed propensity scores over which local polynomials were extrapolated. The shaded region represents 90% confidence intervals, constructed via bootstrap resampling.

Figure 8: Estimated MCPFs for Piece-Rate Taxes



Note: This figure plots estimated marginal costs of public funds (MCPFs) for piece-rate taxes. The green line plots estimated MCPFs associated with hypothetical piece-rate tax rates denoted on the horizontal axis, and the pink line plots estimated MCPFs associated with a marginal increase to the piece-rate tax rate on the horizontal axis. The vertical line denotes the tax that achieves the “efficient” hourly supply share found in a full-information equilibrium. Shaded regions represent 90% confidence intervals, constructed via bootstrap resampling.

Table 1: Experimental Group Assignment

Hourly Wage Offer	Piece-Rate Offer	Number of Participants
No Hourly Offer	\$0.03 per sentence	302
\$1.20/hr	\$0.03 per sentence	300
\$1.80/hr	\$0.03 per sentence	101
\$2.40/hr	\$0.03 per sentence	103
\$3.00/hr	\$0.03 per sentence	304
\$3.60/hr	\$0.03 per sentence	100
\$4.20/hr	\$0.03 per sentence	99
\$4.80/hr	\$0.03 per sentence	101
\$5.40/hr	\$0.03 per sentence	101
\$6.00/hr	\$0.03 per sentence	305
\$7.20/hr	\$0.03 per sentence	100
\$8.40/hr	\$0.03 per sentence	102
\$9.60/hr	\$0.03 per sentence	101
\$10.80/hr	\$0.03 per sentence	100
\$12.00/hr	\$0.03 per sentence	305
\$15.00/hr	\$0.03 per sentence	100
\$18.00/hr	\$0.03 per sentence	102
\$21.00/hr	\$0.03 per sentence	304
		<i>Total:</i> 3030

Note: This table summarizes the treatment conditions and sample sizes for each experimental group in the pilot. *Piece-rate offer* denotes the performance-based bonus offer, which is awarded on a per-sentence basis and common across all experimental groups. *Hourly wage offer* denotes the fixed-rate compensation offered to workers for the five-minute task, prorated to one hour. Note that both piece rate and hourly wage options are offered as supplements to a base wage of \$12.00 per hour.

Table 2: Summary Statistics

Category	Variable	Mean	SD
<i>Panel A: Task Performance</i>	Accepted Hourly Offer	0.486	0.500
	Completed Sentences	21.98	8.143
	Correct Sentences	19.45	8.480
	Output Value	7.003	3.053
	Finished	0.986	0.117
<i>Panel B: Demographics & Employment</i>	Age	37.23	12.18
	Female	0.643	0.479
	Minority	0.357	0.479
	Employed	0.685	0.465
	Student	0.187	0.390
	Number of Previous Tasks	1281.6	1746.4

Note: This table reports summary statistics for the experimental sample. Panel A reports statistics on variables related to experimental task performance and experience. Panel B reports demographic information. The total number of participating workers is 3,030.

Table 3: Logit Estimates of Hourly Supply

	(1)	(2)	(3)	(4)
	Accepted Offer	Accepted Offer	Accepted Offer	Accepted Offer
Log Hourly Wage Offer	1.202*** (0.0555)	1.205*** (0.0555)	1.216*** (0.0561)	1.248*** (0.0579)
Number of Previous Tasks/1000			0.0145 (0.0252)	−0.00270 (0.0261)
Age				0.0292*** (0.00428)
Female				0.281*** (0.0957)
Task Controls	No	Yes	Yes	Yes
Employment Controls	No	No	Yes	Yes
Demographic Controls	No	No	No	Yes
<i>N</i>	2728	2728	2728	2728

Note: This table reports estimated coefficients from logistic regressions of hourly contract acceptance against log wage offers, excluding control-group workers who were only offered a piece rate. Columns (2)–(4) add control variables for the categories of observable characteristics listed in the bottom panel. Task controls include indicators for experimental wave and start time. Employment controls include unemployment and not-in-labor-force indicators, student enrollment status, and number of previous tasks completed on Prolific. Demographic controls include race, gender, and age. *** = significant at 1% level, ** = significant at 5% level, * = significant at 10% level.

Table 4: 2SLS Estimates of Treatment Effects of Hourly Wages on Output Value

	(1) Output Value	(2) Output Value	(3) Output Value	(4) Output Value
Accepted Hourly Offer	-0.612*** (0.215)	-0.606*** (0.208)	-0.591*** (0.207)	-0.475** (0.193)
Number of Previous Tasks/1000			0.178*** (0.0338)	0.178*** (0.0325)
Age				-0.0429*** (0.00446)
Female				0.557*** (0.114)
Task Controls	No	Yes	Yes	Yes
Employment Controls	No	No	Yes	Yes
Demographic Controls	No	No	No	Yes
Dep. Mean	7.00	7.00	7.00	7.00
R-squared	0.040	0.088	0.104	0.221
N	3030	3030	3030	3030

Note: This table reports estimated coefficients from two-stage least-squares regressions of residual output value, where output value equals correctly typed sentences per hour times \$0.03, against an indicator for accepting an hourly wage offer. I partial-out wage effects by regressing output value against treatment offers and log effective hourly wages among hourly workers, then subtracting the demeaned wage effect implied by the coefficient on log effective hourly wages. I then instrument for hourly wage take-up with log wage offer and an indicator variable for being in the no-offer control group. Columns (2)–(4) add control variables for the categories of observable characteristics listed in the bottom panel. Task controls include indicators for experimental wave and start time. Employment controls include unemployment and not-in-labor-force indicators, student enrollment status, and number of previous tasks completed on Prolific. Demographic controls include race, gender, and age. *** = significant at 1% level, ** = significant at 5% level, * = significant at 10% level.

Table 5: OLS Estimates of Selection on Output Value by Wage Offer

	(1) Output Value	(2) Output Value	(3) Output Value	(4) Output Value
Accepted Hourly Offer	-2.664*** (0.340)	-2.552*** (0.333)	-2.513*** (0.334)	-2.387*** (0.312)
Declined $\times$ Log Hourly Wage Offer	0.0927 (0.100)	0.119 (0.0970)	0.141 (0.0965)	0.163* (0.0906)
Accepted $\times$ Log Hourly Wage Offer	0.666*** (0.121)	0.610*** (0.117)	0.605*** (0.117)	0.555*** (0.110)
Accepted $\times$ Log Effective Hourly Wage	-0.109 (0.126)	-0.0860 (0.123)	-0.0815 (0.123)	-0.0410 (0.116)
Control Group (No Hourly Offer)	-0.361 (0.234)	-0.323 (0.226)	-0.263 (0.226)	-0.263 (0.209)
Number of Previous Tasks/1000			0.180*** (0.0326)	0.177*** (0.0313)
Age				-0.0359*** (0.00430)
Female				0.624*** (0.111)
Task Controls	No	Yes	Yes	Yes
Employment Controls	No	No	Yes	Yes
Demographic Controls	No	No	No	Yes
Dep. Mean	7.00	7.00	7.00	7.00
R-squared	0.075	0.121	0.137	0.254
<i>N</i>	3030	3030	3030	3030

Note: This table reports estimated coefficients from OLS regressions of output value (number of correctly typed sentences per hour times \$0.03) against log hourly wage offers interacted with acceptance status, adjusting for log effective wages among hourly workers (see Equation (5) in the text). The coefficient on “Declined $\times$ Log Hourly Wage Offer” captures the change in output value among piece-rate workers for each unit increase in their log hourly wage offer. The coefficient on “Accepted $\times$ Log Hourly Wage Offer” captures the change in output value among hourly workers for each unit increase in their log hourly wage offer. The coefficient on “Accepted $\times$ Log Effective Hourly Wage” captures the change in output value among hourly workers for each unit increase in the log hourly wage they are *paid*, conditional on the wage they are *offered*. Columns (2)–(4) add control variables for the categories of observable characteristics listed in the bottom panel. Task controls include indicators for experimental wave and start time. Employment controls include unemployment and not-in-labor-force indicators, student enrollment status, and number of previous tasks completed on Prolific. Demographic controls include race, gender, and age. *** = significant at 1% level, ** = significant at 5% level, * = significant at 10% level.

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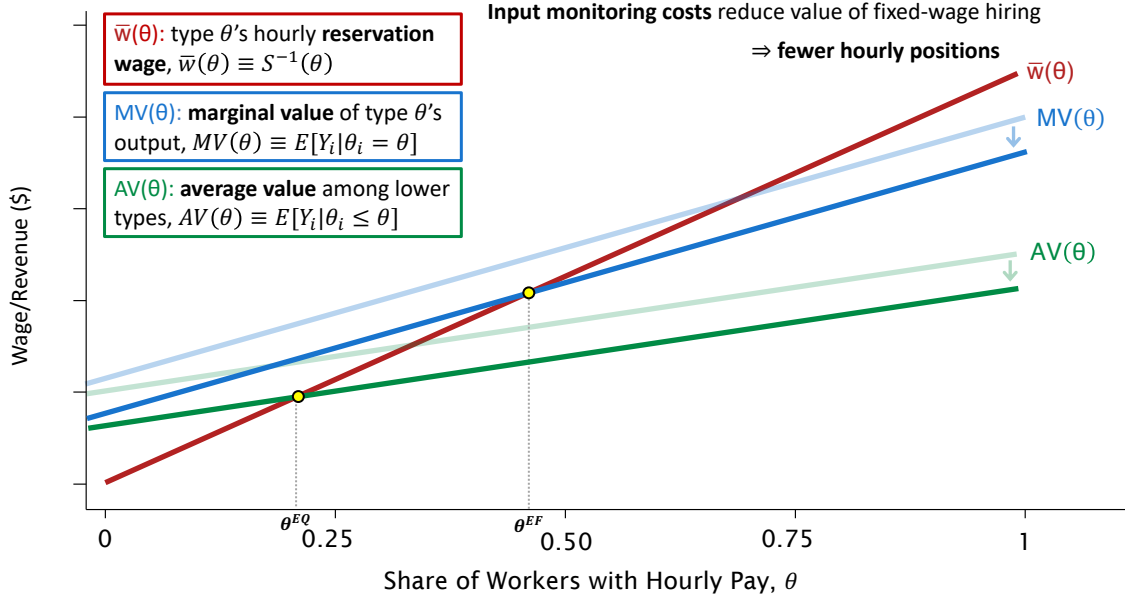
Appendix A Additional Figures and Tables

Figure A1: Test for Bunching at Wage-Offer Reference Points

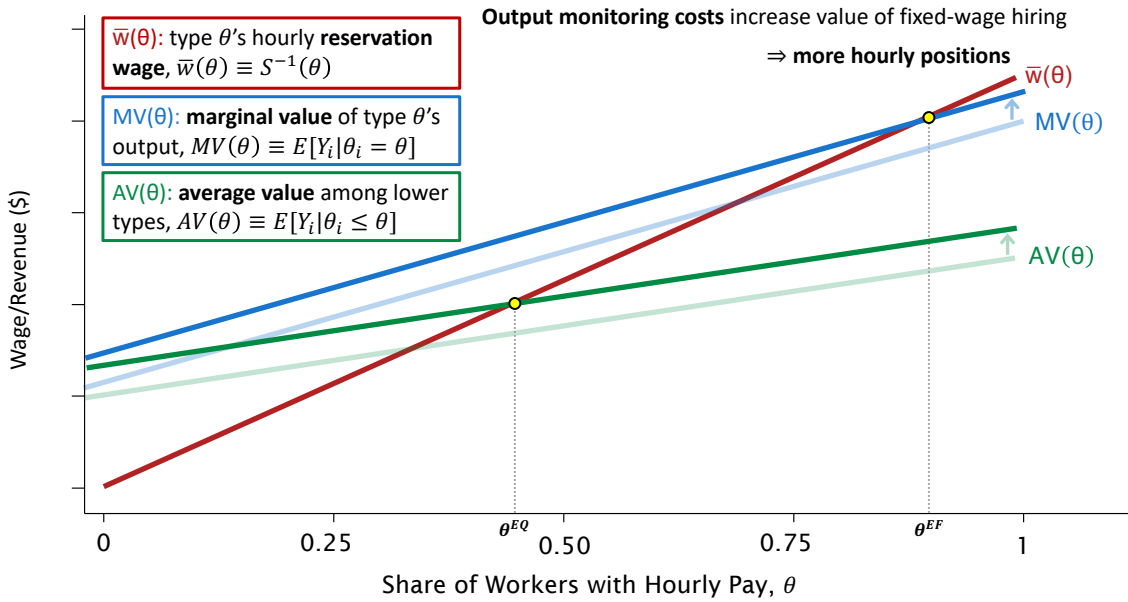


Note: This figure reports estimated densities of realized hourly earnings minus initial wage offers among workers in all treatment-offer groups who declined their wage offers in favor of the piece rate. The red and blue lines denote separate estimated densities for earnings below and above initial wage offers. Second-order local polynomial density estimates are based on a triangular kernel method from Cattaneo et al. (2020), using optimal bandwidths of 3.93 (left) and 3.38 (right) of the cutoff. The p-value for the continuity test at the cutoff is 0.803. The shaded regions indicate 90% confidence intervals computed from bias-corrected jackknife standard errors.

Figure A2: Asymmetric Information with Input and Output Monitoring Costs



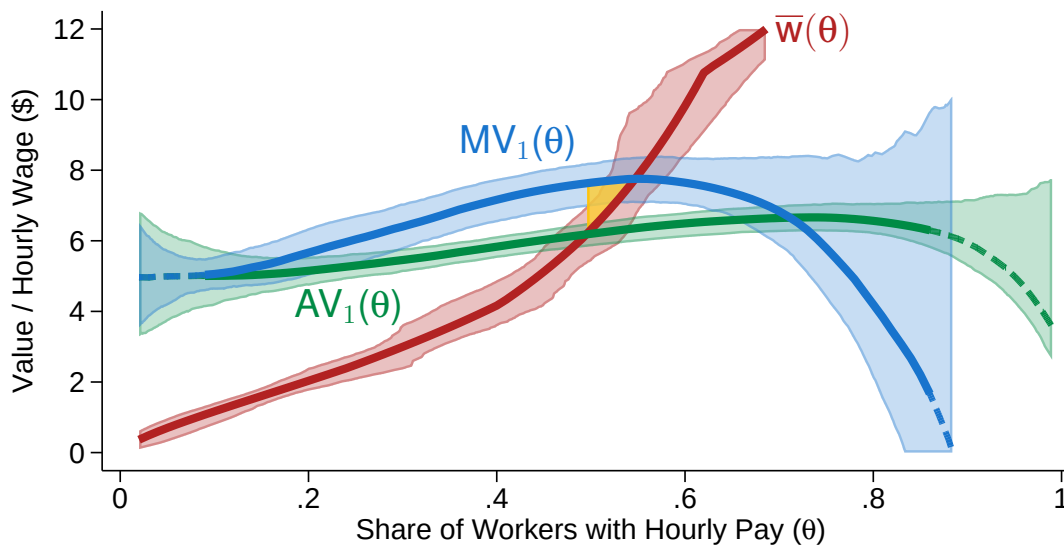
(A) Input Monitoring Costs



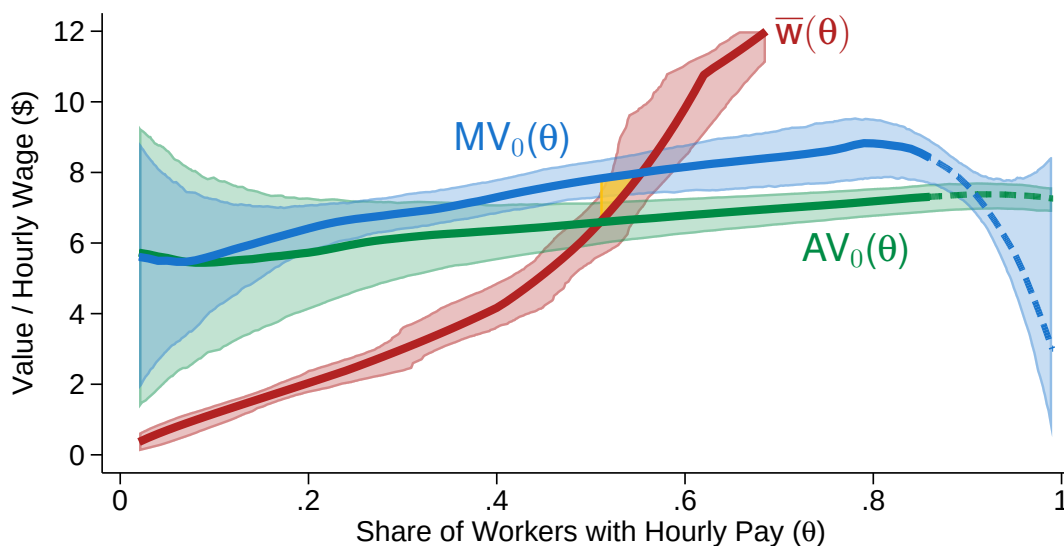
(B) Output Monitoring Costs

Note: This figure provides a graphical representation of a market for hourly wages under asymmetric information with input and output monitoring costs.

Figure A3: Estimates of Marginal and Average Value Curves with Monotone-Spline First Stage



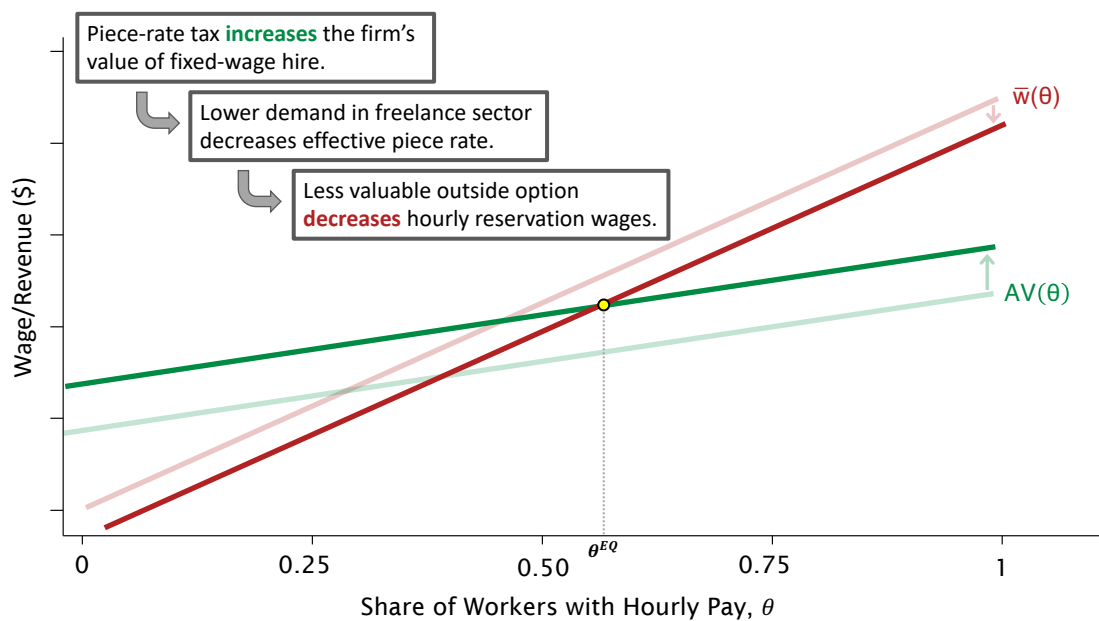
(A) Potential Value Under Hourly Wage



(B) Potential Value Under Piece Rate

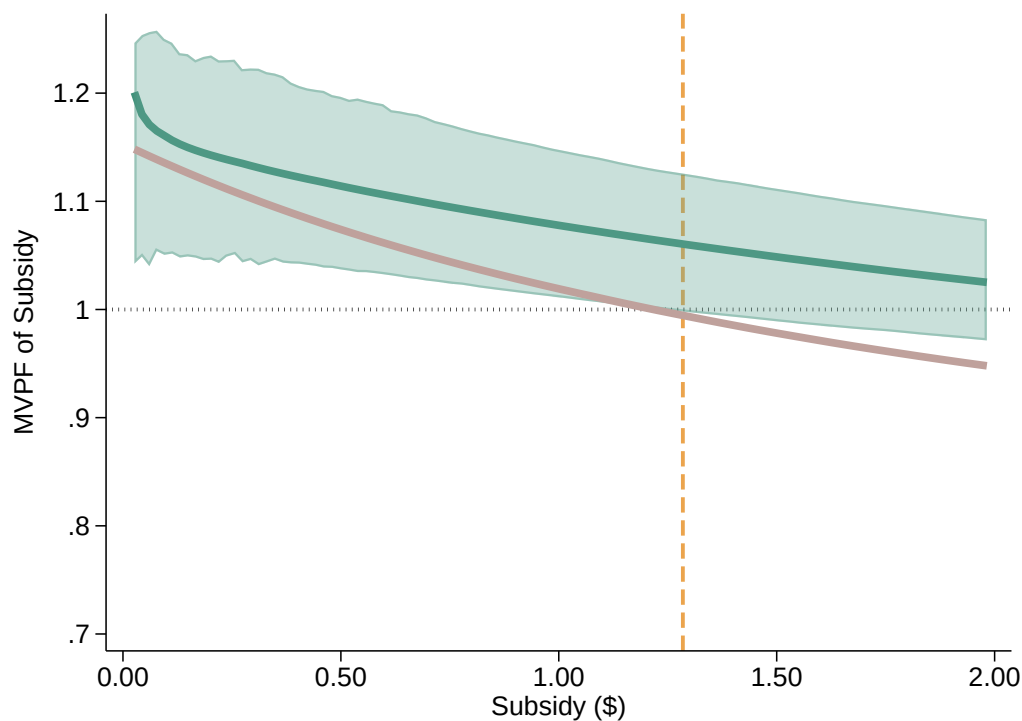
Note: This figure reproduces the value-curve estimates in Figure 6 using an alternative first-stage specification that replaces the logit first-stage propensity score with a monotone increasing spline in log wage offers. Spline estimates of this first-stage supply curve are plotted in red, while blue and green lines plot semi-parametric estimates of marginal value and average value curves under hourly-wage (top panel) and piece-rate (bottom panel) counterfactuals. Areas in yellow denote estimated welfare loss from adverse selection into hourly wages, estimated to be \$0.04 (SE=0.022) and \$0.03 (SE=0.029) per hour of labor, respectively. Shaded regions represent 90% confidence intervals, constructed via bootstrap resampling. Monotone splines are estimated using five interior knots placed at equally spaced quantiles of log wage offers. See Appendix E for estimation details.

Figure A4: Equilibrium Effects of Piece-Rate Taxes



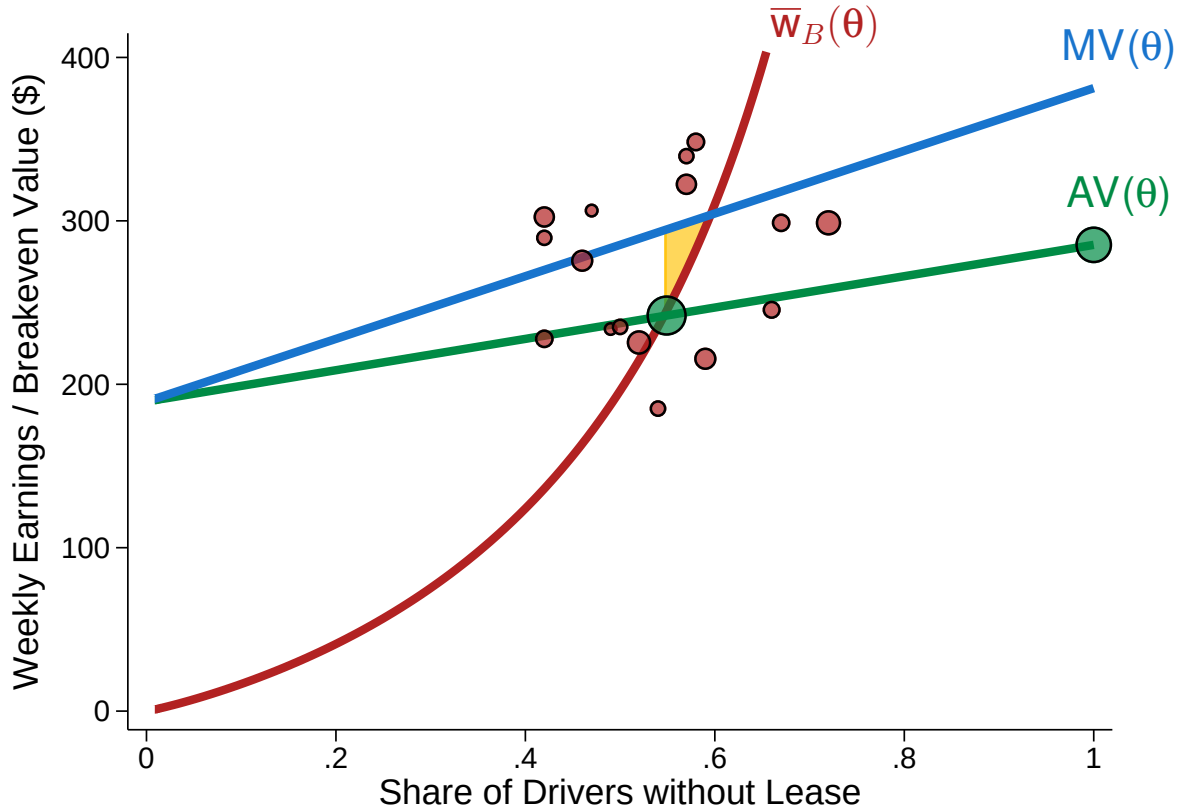
Note: This figure provides a graphical representation of both first- and second-order effects of piece-rate taxes on hourly labor share.

Figure A5: Estimated MVPFs for Hourly-Wage Subsidies



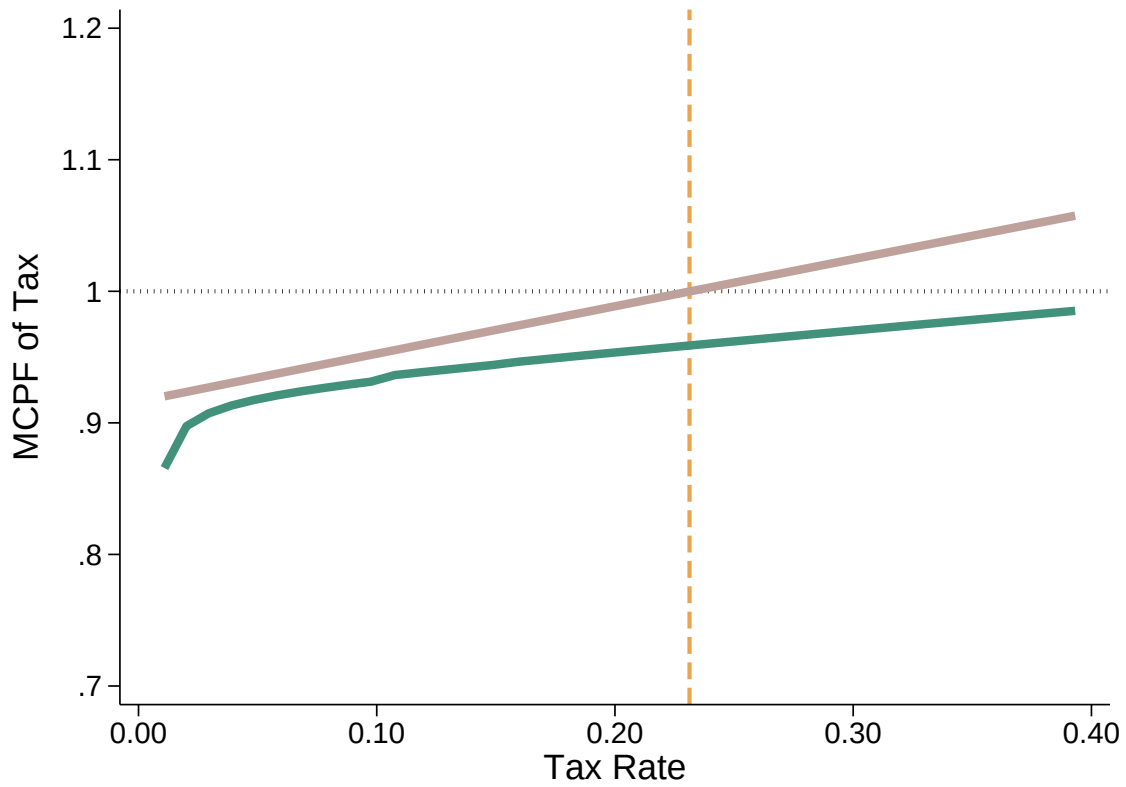
Note: This figure plots estimated marginal values of public funds (MVPFs) for hourly-wage subsidies. The green line plots estimated MVPFs associated with hypothetical hourly-wage subsidies (in dollars per hour worked) denoted on the horizontal axis, and the pink line plots estimated MVPFs associated with a marginal increase to the subsidy. The vertical line denotes the subsidy that achieves the “efficient” hourly supply share found in a full-information equilibrium. MVPFs are constructed as in Appendix F. Shaded regions represent 90% confidence intervals.

Figure A6: Value Curves for Lease-Free Rideshare Driving, Calibrated from Angrist et al. (2021)



Note: This figure plots reservation wages and value curves for lease-free rideshare contracts, calibrated using estimates from Angrist et al. (2021). The horizontal axis denotes the share of drivers declining weekly lease offers. The red curve plots the inverse supply of lease-free drivers, $\bar{w}_B(\theta) \equiv S^{-1}(\theta)$ —the breakeven value, w_B , at which a $1 - \theta$ share of drivers will accept taxi leases. The blue line plots the marginal value curve, $MV(\theta) \equiv E[Y_i | \theta_i = \theta]$, where Y_i denotes driver i 's weekly revenue from ride fares. The green line plots the average value curve, $AV(\theta) \equiv E[Y_i | \theta_i \leq \theta]$. Red circles denote average lease opt-out rates by experimental breakeven offers, residualized by strata- and week-fixed effects. Green circles denote estimates of average fare revenues earned by decliners of lease offers and all drivers, respectively. Circle sizes reflect frequency weights based on reported observation counts. The yellow region corresponds to the welfare loss from adverse selection into lease-free ride share contracts. See Appendix H for details.

Figure A7: Estimated MCPFs for Driver-Fare Taxes for Rideshare Drivers



Note: This figure plots estimated marginal costs of public funds (MCPFs) for a proportional tax on rideshare drivers' fare earnings. The green line plots estimated MCPFs associated with hypothetical driver-fare tax rates denoted on the horizontal axis, and the pink line plots estimated MCPFs associated with a marginal increase to the tax rate on drivers' fare earnings. The vertical line denotes the tax that achieves the "efficient" weekly supply share found in a full-information equilibrium. MCPFs are constructed using calibrated marginal value and supply curves for lease-free rideshare contracts from Angrist et al. (2021), applied to Equation (27) in the text.

Table A1: Balance Test

	(1)	(2)	(3)
	Any Hourly Wage Offer	Experimental Wage Offer	Output Value
Number of Previous Tasks/1000	0.00460 (0.00316)	-0.0446 (0.0718)	0.195*** (0.0321)
Age	0.0000335 (0.000520)	0.00421 (0.0109)	-0.0599*** (0.00477)
Female	-0.00469 (0.0118)	0.156 (0.257)	0.567*** (0.113)
Minority	0.00793 (0.0117)	0.00323 (0.263)	-1.040*** (0.114)
Employed	0.0173 (0.0135)	-0.234 (0.287)	0.144 (0.127)
Student	0.0106 (0.0162)	0.366 (0.355)	-0.538*** (0.156)
F-statistic	0.707	0.872	30.456
<i>p</i> -value	0.733	0.568	0.000
<i>N</i>	3030	2728	3030

Note: This table reports results from a test of balanced treatment for experimental hourly wage offers. Columns 1 and 2 report estimated coefficients from OLS regressions of experimental group assignment measures against the baseline demographic variables reported in the leftmost column. The outcome variable in Column 1 is an indicator for receiving any experimental wage offer. The outcome variable in Column 2 is the level of wage offer received among workers not in the control group. Column 3 reports estimated coefficients from the same specification, but with output value as the dependent variable. The bottom rows report F-statistics and *p*-values from a test of joint significance for all right-hand side variables.

Table A2: 2SLS Estimates of Treatment Effects of Hourly Wages on Output Value without Adjusting for Wage Effects

	(1) Output Value	(2) Output Value	(3) Output Value	(4) Output Value
Accepted Hourly Offer	-0.654*** (0.215)	-0.648*** (0.208)	-0.634*** (0.207)	-0.517*** (0.193)
Number of Previous Tasks/1000			0.178*** (0.0337)	0.178*** (0.0324)
Age				-0.0426*** (0.00445)
Female				0.561*** (0.113)
Task Controls	No	Yes	Yes	Yes
Employment Controls	No	No	Yes	Yes
Demographic Controls	No	No	No	Yes
R-squared	0.042	0.090	0.106	0.223
<i>N</i>	3030	3030	3030	3030

Note: This table reports estimated coefficients from two-stage least-squares regressions of unresidualized output value against an indicator for accepting an hourly wage offer. I instrument for hourly wage take-up with log wage offer and an indicator variable for being in the no-offer control group. Columns (2)–(4) add control variables for the categories of observable characteristics listed in the bottom panel. Task controls include indicators for experimental wave and start time. Employment controls include unemployment and not-in-labor-force indicators, student enrollment status, and number of previous tasks completed on Prolific. Demographic controls include race, gender, and age. *** = significant at 1% level, ** = significant at 5% level, * = significant at 10% level.

Table A3: 2SLS Estimates of Treatment Effects of Hourly Wages on Output Value Based on Completed Sentences

	(1)	(2)	(3)	(4)
	Output Value	Output Value	Output Value	Output Value
Accepted Hourly Offer	-0.503** (0.206)	-0.498** (0.200)	-0.485** (0.199)	-0.362** (0.185)
Number of Previous Tasks/1000			0.164*** (0.0338)	0.174*** (0.0322)
Age				-0.0528*** (0.00420)
Female				0.365*** (0.108)
Task Controls	No	Yes	Yes	Yes
Employment Controls	No	No	Yes	Yes
Demographic Controls	No	No	No	Yes
Dep. Mean	7.91	7.91	7.91	7.91
R-squared	0.037	0.080	0.095	0.232
<i>N</i>	3030	3030	3030	3030

Note: This table reports estimated coefficients from two-stage least-squares regressions of residual output value, where output value equals completed sentences per hour times \$0.03, against an indicator for accepting an hourly wage offer. I partial-out wage effects by regressing output value against treatment offers and log effective hourly wages among hourly workers, then subtracting the demeaned wage effect implied by the coefficient on log effective hourly wages. I then instrument for hourly wage take-up with log wage offer and an indicator variable for being in the no-offer control group. Columns (2)–(4) add control variables for the categories of observable characteristics listed in the bottom panel. Task controls include indicators for experimental wave and start time. Employment controls include unemployment and not-in-labor-force indicators, student enrollment status, and number of previous tasks completed on Prolific. Demographic controls include race, gender, and age. *** = significant at 1% level, ** = significant at 5% level, * = significant at 10% level.

Table A4: OLS Estimates of Selection on Output Value Based on Completed Sentences by Wage Offer

	(1) Output Value	(2) Output Value	(3) Output Value	(4) Output Value
Accepted Hourly Offer	-2.522*** (0.325)	-2.425*** (0.318)	-2.383*** (0.318)	-2.195*** (0.297)
Declined $\times$ Log Hourly Wage Offer	0.172* (0.0961)	0.196** (0.0932)	0.213** (0.0925)	0.233*** (0.0855)
Accepted $\times$ Log Hourly Wage Offer	0.616*** (0.116)	0.565*** (0.112)	0.564*** (0.113)	0.497*** (0.104)
Accepted $\times$ Log Effective Hourly Wage	-0.0796 (0.121)	-0.0583 (0.117)	-0.0583 (0.118)	-0.0153 (0.110)
Control Group (No Hourly Offer)	-0.330 (0.226)	-0.296 (0.219)	-0.242 (0.219)	-0.228 (0.201)
Number of Previous Tasks/1000			0.166*** (0.0325)	0.173*** (0.0309)
Age				-0.0454*** (0.00402)
Female				0.435*** (0.105)
Task Controls	No	Yes	Yes	Yes
Employment Controls	No	No	Yes	Yes
Demographic Controls	No	No	No	Yes
Dep. Mean	7.91	7.91	7.91	7.91
R-squared	0.081	0.122	0.138	0.272
<i>N</i>	3030	3030	3030	3030

Note: This table reports estimated coefficients from OLS regressions of output value (number of correctly completed sentences per hour times \$0.03) against log hourly wage offers interacted with acceptance status, adjusting for log effective wages among hourly workers (see Equation (5) in the text). The coefficient on “Declined $\times$ Log Hourly Wage Offer” captures the change in output value among piece-rate workers for each unit increase in their log hourly wage offer. The coefficient on “Accepted $\times$ Log Hourly Wage Offer” captures the change in output value among hourly workers for each unit increase in their log hourly wage offer. The coefficient on “Accepted $\times$ Log Effective Hourly Wage” captures the change in output value among hourly workers for each unit increase in the log hourly wage they are *paid*, conditional on the wage they are *offered*. Columns (2)–(4) add control variables for the categories of observable characteristics listed in the bottom panel. Task controls include indicators for experimental wave and start time. Employment controls include unemployment and not-in-labor-force indicators, student enrollment status, and number of previous tasks completed on Prolific. Demographic controls include race, gender, and age. *** = significant at 1% level, ** = significant at 5% level, * = significant at 10% level.

Table A5: Multiple Hypothesis Tests

	Multiplicity-Adjusted p -values			
Selection on Y_0				
$H_0 : \beta_0 = 0$	0.3483	0.2303	0.1527	0.0767
Selection on Y_1				
$H_0 : \beta_1 = 0$	0.0003	0.0003	0.0003	0.0003
Treatment Effect				
$H_0 : \psi = 0$	0.0037	0.0030	0.0037	0.0183
Task Controls	No	Yes	Yes	Yes
Employment Controls	No	No	Yes	Yes
Demographic Controls	No	No	No	Yes

Note: This table reports p -values adjusted for multiple hypothesis testing following Theorem 3.1 from List et al. (2019). Each cell reports the smallest family-wise error rate (the rate of at least one false rejection) across all three hypotheses that still rejects the null hypothesis listed in that row. The first row corresponds to the null hypothesis of zero selection into higher wage offers on potential output under the piece rate ($H_0 : \beta_0 = 0$ in Equation (5)). The second row corresponds to the null hypothesis of zero selection into higher wage offers on potential output under the hourly wage ($H_0 : \beta_1 = 0$ in Equation (5)). The third row corresponds to the null hypothesis of a zero reduced-form effect of the wage offer on output ($H_0 : \psi = 0$, where $\psi = Cov(Y_i, \widetilde{W}_i)/Var(\widetilde{W}_i)$ and $\widetilde{W}_i$ is wage offer residualized on controls). Columns (2)–(4) add control variables for the categories of observable characteristics listed in the bottom panel. Task controls include indicators for experimental wave and start time. Employment controls include unemployment and not-in-labor-force indicators, student enrollment status, and number of previous tasks completed on Prolific. Demographic controls include race, gender, and age.

Appendix B Identification of Experimental Estimands

This appendix provides additional details and proofs behind the target estimands in my experimental design.

B.1 Identifying Average Selection in Multiple-Offer Experiment

Consider an example experiment with three offer conditions, $W_i \in \{0, L, H\}$. As in the previous example, control workers assigned to $W_i = 0$ are offered the piece rate with no alternative. But now the remaining workers are randomly separated into two groups—workers assigned to $W_i = L$ are offered the choice between the piece rate and low hourly wage, while workers assigned to $W_i = H$ are offered the choice between the piece rate and a high hourly wage. Let D_i^L and D_i^H be indicator for individual i 's potential take-up of contracts L and H , respectively, and assume $D_i^H \geq D_i^L$ for all i .

As in Equation (3), comparing decliners of a given wage offer with control workers identifies average selection on Y_0 into that offer among all workers. But now I can also compare outcomes between decliners of high- and low-offer treatment offers to identify selection on Y_0 into offer H among those who would reject the less generous offer (L):

$$\begin{aligned} & \underbrace{E[Y_{0i} | D_i^H = 1, D_i^L = 0] - E[Y_{0i} | D_i^H = 0]}_{\text{Average Selection on } Y_0} \\ &= \frac{1 - \pi^L}{\pi^H - \pi^L} (E[Y_i | D_i = 0, W_i = L] - E[Y_i | D_i = 0, W_i = H]). \end{aligned} \tag{30}$$

At the same time, a comparison between *accepters* of high- and low-offer treatment offers identifies average selection on Y_1 into offer L among those who would accept

the more generous offer (H):

$$\begin{aligned}
& \underbrace{E[Y_{1i}|D_i^L = 1] - E[Y_{1i}|D_i^H = 1, D_i^L = 0]}_{\text{Average Selection on } Y_1} \\
&= \frac{\pi^H}{\pi^H - \pi^L} (E[Y_i|D_i = 1, W_i = L] - E[Y_i|D_i = 1, W_i = H]),
\end{aligned} \tag{31}$$

In short, because both high- and low-offer treatment arms contain a mix of hourly and piece-rate workers, this multiple treatment design allows me to identify worker selection on *both* potential outcomes—productivity under the piece rate (Y_0) and productivity under hourly wages (Y_1).

B.2 Identification of Wage Effects

So far, I have assumed that a worker’s assigned offer condition can only affect their outcome through the choice of hourly versus piece-rate contract, $W_i \perp\!\!\!\perp (Y_{1i}, Y_{0i})$. If hourly workers are paid their offered wages, this exclusion restriction could be violated through wage effects—higher pay might induce greater effort through increased motivation or satisfaction, biasing my estimates of both selection and treatment effects.

To separate the potential behavioral response of higher effective wages from the incentive effects of hourly contract structure, my experiment incorporates an additional dimension of randomization in the spirit of Karlan and Zinman (2009). Specifically, after workers choose their compensation option, but before they begin the task, I increase hourly wages for a random subset of those accepting lower wage offers, bringing them to parity with higher treatment-offer groups.

To see how randomized wage top-ups separately identifies wage effects and selection on Y_1 , consider an individual i who receives a job offer, W_i , at one of two randomized wages: a high offer ($W_i = H$) or a low offer ($W_i = L$). Let D_{W_i} denote the individual’s potential acceptance of a offer W , so that $D_{Hi} = 1$ if i would accept the high offer and $D_{Li} = 1$ if i would accept the low offer. Furthermore, let Y_{Hi} and Y_{Li} denote the potential output levels produced by i if they were paid hourly wages of H and L , respectively. Note that if realized wages reflected accepted offers, comparing output between those who accept H and those who accept L would yield

the following:

$$\begin{aligned}
& E[Y_i|W_i = H, D_i = 1] - E[Y_i|W_i = L, D_i = 1] \\
&= \underbrace{E[Y_{Hi} - Y_{Li}|D_{Li} = 1]}_{\text{Wage Effect}} + \underbrace{E[Y_{Hi}|D_{Hi} = 1] - E[Y_{Hi}|D_{Li} = 1]}_{\text{Selection}}. \tag{32}
\end{aligned}$$

This difference is the sum of both the wage effect and selection of H relative to L , which cannot be separated without observing $E[Y_{Hi}|D_{Li} = 1]$.

Now let W_i^P be an indicator whether individual i receives a surprise wage increase of $\Delta = H - L$ after accepting their contract. W_i^P is randomly assigned among those who received low offers ($W_i = L$) and accepted them ($D_{Li} = 1$) but is zero for everyone else. With this randomized wage raise, I can estimate wage effects by comparing output between low- and high-wage workers in the low-offer group:

$$\begin{aligned}
\text{Wage Effect} &= E[Y_i|W_i = L, D_i = 1, W_i^P = 1] - E[Y_i|W_i = L, D_i = 1, W_i^P = 0] \\
&= E[Y_{Hi} - Y_{Li}|D_{Li} = 1] \tag{33}
\end{aligned}$$

And I can estimate selection by comparing output between low- and high-offer groups with high realized wages:

$$\begin{aligned}
\text{Selection} &= E[Y_i|W_i = H, D_{Hi} = 1] - E[Y_i|W_i = L, D_{Li} = 1, W_i^P = 1] \\
&= E[Y_{Hi}|D_{Hi} = 1] - E[Y_{Hi}|D_{Li} = 1]. \tag{34}
\end{aligned}$$

B.3 Identifying Marginal Value in a Linear Model

Drawing from Equations (30) and (31), consider the average potential outcomes among workers who reject over L but accept offer H .

$$\begin{aligned}
E[Y_{1i}|D_i^H = 1, D_i^L = 0] &= \frac{\pi^H E[Y_i|D_i = 1, W_i = H] - \pi^L E[Y_i|D_i = 1, W_i = L]}{\pi^H - \pi^L} \tag{35} \\
E[Y_{0i}|D_i^H = 1, D_i^L = 0] &= \frac{(1 - \pi^L) E[Y_i|D_i = 0, W_i = L] - (1 - \pi^H) E[Y_i|D_i = 0, W_i = H]}{\pi^H - \pi^L} \tag{36}
\end{aligned}$$

Let $\bar{w}_i$ denote the lowest offer individual i is willing to accept. Let $H = w$ and $L = w - \tau$ in Equations (35) and (36). The limits of $E[Y_{1i}|D_i^H = 1, D_i^L = 0]$ and

$E[Y_{0i}|D_i^H = 1, D_i^L = 0]$ as $\tau \rightarrow 0$ can be written as

$$E[Y_{1i}|\bar{w}_i = w] = \frac{\partial(E[Y_i|D_i(w) = 1]S(w))}{\partial S(w)} \quad (37)$$

$$E[Y_{0i}|\bar{w}_i = w] = -\frac{\partial(E[Y_i|D_i(w) = 0](1 - S(w)))}{\partial S(w)}. \quad (38)$$

Now suppose both $E[Y|D_i(w) = 1]$, $E[Y|D_i(w) = 0]$, and $S(w) \equiv \Pr(\bar{w}_i \leq w)$, are all linear in the wage offer, w :

$$S(w) = \alpha + \beta w \quad (39)$$

$$E[Y_i|D_i(w) = 1] = \gamma_1 + \delta_1 w \quad (40)$$

$$E[Y_i|D_i(w) = 0] = \gamma_0 + \delta_0 w. \quad (41)$$

We therefore have

$$E[Y_{1i}|\bar{w}_i = w] = \frac{(\gamma_1 + \delta_1 w)\beta + (\alpha + \beta w)\delta_1}{\beta} \quad (42)$$

$$= \frac{\alpha\delta_1}{\beta} + \gamma_1 + 2\delta_1 w. \quad (43)$$

Likewise for $E[Y_{0i}|\bar{w}_i = w]$:

$$E[Y_{0i}|\bar{w}_i = w] = \frac{-(\gamma_0 + \delta_0 w)\beta + (1 - \alpha - \beta w)\delta_0}{-\beta} \quad (44)$$

$$= \frac{(\alpha - 1)\delta_0}{\beta} + \gamma_0 + 2\delta_0 w. \quad (45)$$

We therefore have

$$\frac{\partial E[Y_{1i}|\bar{w}_i = w]}{\partial w} = 2\delta_1 \quad (46)$$

$$\frac{\partial E[Y_{0i}|\bar{w}_i = w]}{\partial w} = 2\delta_0 \quad (47)$$

Appendix C Experimental Protocol

The design and recruitment details for this experiment were pre-registered on the AEA RCT Registry under ID AEARCTR-0000714, titled “Asymmetric Information in Labor Contracts: Evidence from an Online Experiment” (Herbst, 2024). The experiment took place in ten waves of three-hundred job postings launched over the course of two weeks beginning August 31, 2024. Figure C1 provides a timeline of the experimental protocol.

Participants in my experiment were recruited on Prolific, an online platform that allows clients to hire online workers for short-term tasks.⁴⁶ The experimental job posting offered workers a \$1.00 reward for transcribing handwritten text into typed form for five minutes. Such transcription tasks are commonly requested on Prolific and other online platforms, often for the purpose of training artificial intelligence (AI) algorithms. The posting also informed workers they “can earn an additional \$0.03 in bonus compensation for each correctly typed sentence.”

Workers could only see my experimental job posting if they met the following screening criteria: (1) were located in the United States, (2) spoke fluent English, (3) successfully completed ten or more previous tasks, and (4) earned an approval rate above 98 percent on previous tasks.⁴⁷ These screening criteria allow me to remove the small number of casual users who may take the tasks less seriously than “professional” online workers who regularly perform tasks to earn income. In doing so, they make the sample more representative of the online hiring pool faced by profit-conscious employers.

Workers who accept the job posting are provided with a URL link to the experimental task. Upon clicking this link, workers are shown a screen with a brief task description and example entry.⁴⁸ After clicking past the description page, workers are randomized into one of eighteen experimental groups. Each group is offered a different menu of compensation options in exchange for completing the five-minute data-entry task. In the first treatment group, workers are offered a choice between a fixed \$0.10 payment (\$1.20 per hour) or a piece rate of \$0.03 per correctly typed sen-

⁴⁶Douglas et al. (2023) finds that the Prolific platform compares favorably to Amazon Mechanical Turk (“MTurk”) and other platforms across several dimensions of data quality.

⁴⁷More than 95 percent of Prolific workers meet the 98-percent approval threshold.

⁴⁸The task is hosted on the Qualtrics platform. Readers can view and perform a replication of the task [here](#). Screenshots are provided in Figure C2.

tence.⁴⁹ In the second treatment group, workers are offered a choice between a fixed \$0.15 payment (\$1.80 per hour) or the same \$0.03 piece rate. Additional treatment groups follow the same structure, with each condition offering the \$0.03 piece rate but increasing the flat wage offer by multiples of \$0.05, up to a maximum of \$1.75 (\$21.00 per hour). A control group is offered the \$0.03 piece rate for each correctly typed sentence, with no alternative option. Each of these options is offered as an addition to the \$1.00 reward advertised in the job posting, which all workers receive for agreeing to the task. Experimental conditions are summarized in Table 1.

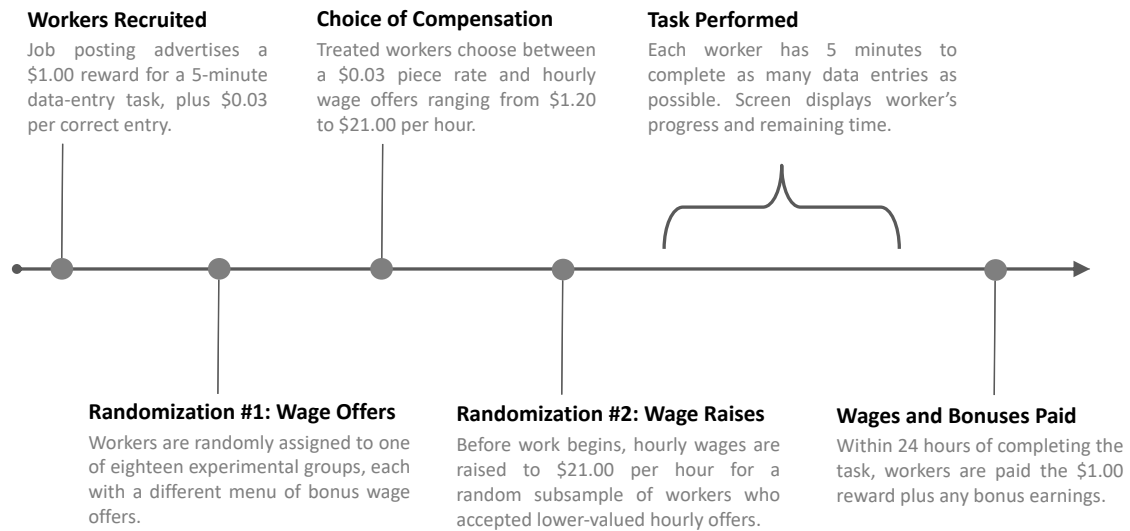
After receiving detailed instructions for the data-entry task, treated workers are presented with their group’s payment options in randomized order, as shown in Figure C2, Panel B. Once workers choose their compensation scheme, they are brought to a new page that states, “For performing this task, you will receive \$1.00, plus your chosen bonus of [*payment choice*].” A random 50 percent of workers who choose lower-valued payment options receive a modified message that increases their base payment by enough to equalize their total compensation with the most generous offer ($\$1.00 + \$1.75 = \$2.75$). For example, half of those who select the \$0.25 payment are told “you will receive \$2.50, plus your chosen bonus of \$0.25.”

Once workers are notified of their compensation and click “Begin Task,” they are presented with a handwritten sentence and a text box. The worker types a sentence in the box and clicks the “Next” button, bringing them to a new page with a different sentence. This process continues for five minutes. Worker output is validated in real time, so workers can see a running tally of their score (the number of correctly typed sentences) and their total earnings in the lower-left corner of each page. Workers also see a countdown timer displaying the number of minutes and seconds remaining in the task.⁵⁰ When the timer reaches zero, the screen refreshes to an end-of-task page displaying a performance summary and a completion link to redeem their earnings. Workers are paid the \$1.00 reward plus any bonus earnings within 24 hours of completing the task.

⁴⁹A piece rate of \$0.03 per sentence was chosen to roughly align with the market rate for online text-to-text transcription services (GMR Transcription, 2024; GoTranscript, 2024; Ditto Transcripts, 2024; Transcription Services, 2024).

⁵⁰Figure C2, Panel C provides a screenshot of the task. The display and submission methods for this task were designed to prevent workers from cheating through automation software or bots. While it is possible that some workers may have tried to make use of such software, performance statistics suggest any such attempts were unsuccessful at increasing output—the maximum score achieved was 52.

Figure C1: Experiment Timeline



Note: This figure provides a timeline for a single wave of the experiment.

Figure C2: Example Job Posting

You will be shown a series of handwritten sentences. On each page, your task is to type the sentence into the text box below.

Here is one example of a completed sentence:

The quick brown fox jumps over the lazy dog.

The quick brown fox jumps over the lazy dog.

Your answers should be as accurate as possible. Please be mindful of capitalization, spacing, and punctuation. When you've completed a sentence, click the "→" button to move on.

You will have 5 minutes to complete as many sentences as you can. You cannot start over, and you can only perform this task once.

(A) Task Description

Before you begin the task, we'd like to offer you a choice of how to earn your bonus payment. Please select your preferred bonus compensation from the options below:

Get paid a \$0.03 bonus for each sentence you correctly complete. ☐

Get paid a flat bonus of \$1.00. ☐

(B) Example Wage Offer

Time Remaining: 03:02

The car sped down the winding country road.

The car sped down the windi

Score: 7
Earnings: \$1.21

→

(C) Typing Task

Note: This figure provides screenshots of the experimental intervention. Panel A shows the task description workers see before they see their wage offer. Panel B shows an example wage offer workers see before they begin the task. Panel C shows the sentence-typing task while it is being performed.

Appendix D Model Extensions

My model is designed so that objects of interest can be mapped to semi-parametric estimands from my experiment. For this reason, it omits features like monitoring costs and dynamic wage-setting, which are absent from my experimental setting and many other short-term labor markets. It also omits wage effects, which are statistically insignificant in my setting (see Section 3). In this appendix, I discuss how one might extend the model to incorporate these features.

Monitoring Costs Existing research shows how relative costs of monitoring worker inputs and outputs can influence worker productivity and equilibrium wage structure in a variety of occupations (Lazear, 1986, 2000; Goldin, 1986; Nagin et al., 2002). My paper seeks to complement this literature by identifying the market implications of asymmetric information holding these monitoring costs fixed. As such, worker time and productivity is costlessly observed in both my model and experimental setting. Nonetheless, one could easily extend my framework to incorporate a monitoring cost of measuring worker output, q_i . Such output monitoring costs would, all else equal, make fixed wage contracts more likely than payment schemes that require precise measurement of individual worker productivity.⁵¹ Likewise, I could allow firms to face an input-monitoring cost of observing workers’ time spent on the job, which would make fixed wages less likely than output-based pay. Appendix Figure A2 shows modified versions of my model that incorporate monitoring costs for inputs and outputs, respectively.

Mixed Compensation In my baseline model, fixed-wage contracts transfer the entirety of a worker’s labor product, Y_i , to the employer. In reality, a firm could offer partial contracts that transfer some share of that labor product. For example, a firm might combine hourly wages with a smaller piece-rate portion to ensure workers have some “skin in the game,” similar to restaurant tipping or sales commissions.

My model can easily be recast to consider such partial contracts. Imagine firms offer a fixed hourly payment, w^p , in exchange for some *portion*, $\eta \leq 1$, of workers’ output value, $Y_i^p \equiv \eta Y_i$. Contracted workers would receive this fixed hourly payment or “base wage,” w^p , in addition to a variable piece-rate component, $(1 - \eta)Y_i$, corre-

⁵¹Note that fixed wages still require some degree of output or effort monitoring so that firms can credibly threaten low-performing workers with dismissal, rejection, or damaged reputation.

sponding to the portion of their labor product not covered by the contract. It is not difficult to see how, much like an insurance co-payment, this piece-rate component would attenuate moral hazard effects from the (partial) fixed wage—as η decreases, the worker bears more of the cost of decreased effort, lowering their incentive to shirk.

Wage Effects The theoretical framework in Section 4 allows workers’ expected output to vary between hourly versus piece-rate compensation. It does not, however, allow that output to vary with the wage level under an hourly contract. In other words, it ignores any potential wage effects that higher hourly compensation might have on worker output. While the absence of wage effects in my empirical results would seem to validate this assumption, I can incorporate wage effects into the model by allowing each worker’s potential output under the hourly contract to vary with the wage (i.e., $Y_{1i} = Y_{1i}(w)$). With this added dimension to potential outcomes, I rewrite $AV_1(\theta)$ as the average value of output among lower types *at θ ’s reservation wage*:

$$AV_1^E(\theta) \equiv E[Y_{1i}(\bar{w}(\theta)) | \theta_i \leq \theta]. \quad (48)$$

Assuming wage effects are weakly positive and non-decreasing in θ , the equilibrium condition is given by $\bar{w}(\theta^{EQ}) = AV_1^E(\theta^{EQ})$. In this case, firms pay an hourly wage equal to the average value of accepting workers’ output *under that wage*, $AV^E(\theta^{EQ})$. Relative to the benchmark model, positive wage effects will therefore push the average value curve upwards and increase the share of hourly contracts under asymmetric information.

Note, however, that the efficient equilibrium—the one that would exist in a full-information counterfactual—is also complicated by the presence of wage effects. A fully-informed firm may benefit from paying a worker above their reservation wage if their expected increase in output exceeds the wage premium (i.e., if $E[Y_{1i}(w) - Y_{1i}(\bar{w}_i) | \theta_i = \theta] > w - \bar{w}(\theta)$ for some w).⁵² I thus rewrite $MV_1(\theta)$ as the marginal value of type θ ’s output *at their profit-maximizing wage*, so

$$MV_1^E(\theta) \equiv E[Y_{1i}(w^*(\theta)) | \theta_i = \theta], \quad (49)$$

⁵²I avoid the term “efficiency wages,” which refers to a class of models explaining unemployment as a general-equilibrium consequence of firms’ strategic wage-setting behavior (Weiss, 2014; Krueger and Summers, 1988; Yellen, 1984). In many efficiency-wage models, above-market wages are driven not by causal effects of wages on productivity, but by worker selection, firms’ monitoring ability, or turnover costs (Salop, 1979; Weiss, 1980).

where

$$w^*(\theta) \equiv \operatorname{argmax}_w E[Y_{1i}(w) - w|\theta_i = \theta]. \quad (50)$$

Note that allowing for wage effects means I can no longer interpret Equation 16 as the marginal treatment effect of hourly-contract take-up—if the wage level influences worker output independently of the hourly compensation structure, the wage-offer instrument no longer satisfies the exclusion restriction. The randomized wage raises in my experimental design eliminate this concern. By equalizing the paid wages of low-offer accepters with those of high-offer accepters, these surprise wage increases isolate variation in *offered* wages conditional on a given *effective* wage. I can therefore identify the marginal treatment effect of being paid a given hourly wage among those indifferent to a particular wage offer. I discuss this instrument validity and estimation of wage effects in Section 2.2.

Dynamic Contracts In dynamic labor markets, workers tend to reveal information about their productivity over time. In long-term employment relationships, employers might use repeated observations of worker output to inform promotions, wage cuts, or dismissals, effectively updating wage contracts to better align each worker’s compensation with their latent productivity and effort (Farber and Gibbons, 1996). In short-term gig markets like my experimental setting, this dynamic screening process operates primarily through work histories, ratings, and reviews, which have been shown to narrow the informational gap between workers and firms (Pallais, 2014; Pallais and Sands, 2016).⁵³ Rather than explicitly incorporate this dynamic evolution of wage contracts, my model considers contracts for a single realization of labor output. Recall, however, that workers in the model are assumed to be pre-screened on observable characteristics. Contracts can be placed in a dynamic context by simply defining this screenable information set to include prior realizations of output or observable measures of reputation. Indeed, my empirical analysis conditions on all information available to firms in my experimental setting, including prior work history (see Footnote 22).

⁵³While informative to firms, these measures of worker experience and reputation are likely less predictive than the repeated instances of past worker performance observed in long-term employment relationships. These relatively limited information sets make short-term employers more vulnerable to adverse selection and moral hazard, perhaps explaining the predominance of output-based pay observed among gig workers.

Appendix E Value Curve Estimation Details

This appendix provides details for estimating the hourly supply and value curves described in Section 5.

To estimate hourly supply as a function of hourly wage offers, my baseline specification uses the logistic regressions in Section 3. This specification ensures that estimates of θ are bound between zero and one and prevents negative reservation wages among low- θ workers. Next, I use the local polynomial regression approach from Carneiro et al. (2011) to estimate average and marginal values. First, I residualize covariates from Y_i separately for hourly and piece-rate workers using double-residual regression methods (Robinson, 1988), assuming these covariates are additively separable from $MV_1(\theta)$ and $MV_0(\theta)$.⁵⁴ To simulate potential screening or (legal) wage discrimination by hypothetical employers, these covariates include controls for number of previous tasks, task start time, and employment status.⁵⁵ For hourly workers, I also include the effective wage paid after any randomized top-up payments in the second round of my experiment. As in Section 3, this residualization prevents potential wage effects from violating the exclusion restriction for the wage-offer instrument. I then estimate marginal and average values using local polynomial regression of residualized Y_i on $S(W_i)$ with a bandwidth of 0.2.⁵⁶ Standard errors are calculated using 1,000 bootstrap replications.

With semi-parametric estimates of $\bar{w}(\theta)$, $MV_1(\theta)$, $AV_1(\theta)$, $MV_0(\theta)$, and $AV_0(\theta)$ curves in hand, it is straightforward to calculate the welfare loss from Equation (13). First, I calculate equilibrium (θ^{EQ}) and efficient (θ^{EF}) shares of hourly wages using the intersection of $\bar{w}(\theta)$ with $AV_1(\theta)$ and $MV_1(\theta)$, respectively. Then, I calculate the cumulative difference in $\bar{w}(\theta)$ and $MV_1(\theta)$ over the region $\theta \in (\theta^{EQ}, \theta^{EF})$. This calculation measures lost welfare as the expected excess output value that piece-rate workers would be willing to forfeit to their employers under the hourly wage contract

⁵⁴More formally, I assume $E[Y_{Ji}|X_i = x, \theta_i = \theta] = \xi_J \tilde{X}_i + MV_J(\theta)$ for $J \in \{0, 1\}$, where $\tilde{X}_i$ is a vector of covariates normalized to mean zero. In other words, X_i can affect the levels of $MV_1(\theta)$ and $MV_0(\theta)$, but not their slopes.

⁵⁵Race, gender, and age were excluded because employers cannot legally use these characteristics in employment or wage-setting decisions. Estimates of linear selection effects from Table 5 suggest including these demographic controls would have minimal effect on the slopes of value curves.

⁵⁶The no-offer control group allows for identification of $AV_0(\theta)$ at infinitely high reservation wages ($\theta \rightarrow 1$). However, because take-up rates fall below one for even the most generous fixed-wage offers, estimating right tails of the value-curve distributions requires some extrapolation of local polynomials. This extrapolation has no meaningful effect on welfare estimates because equilibrium and efficient allocations fall in the middle of the θ distribution.

(see Figure 5, Panel A).

Because $MV_1(\theta)$ and $AV_1(\theta)$ are derived from potential outcomes under the hourly wage, Y_1 , the corresponding welfare calculation includes moral hazard effects. To estimate potential welfare loss without moral hazard effects, I repeat this calculation using piece-rate value curves, $MV_0(\theta)$ and $AV_0(\theta)$.

Alternative Specification with First-Stage Monotone Spline

The baseline semi-parametric specification estimates the first-stage propensity score using a logit model. As a robustness check, I also estimate a more flexible first stage that allows the wage-offer component of the propensity score to vary semi-parametrically, subject to the monotonicity restriction that fixed-wage take-up is weakly increasing in the offered wage.

Let D_i indicate take-up of the hourly contract, let $w_i \equiv \log(W_i)$ denote the log hourly wage offer, and let X_i denote the first-stage covariates. Among workers who receive a positive hourly wage offer, the monotone-spline first stage is

$$\Pr(D_i = 1 \mid w_i, X_i) = \Lambda(\alpha_0 + X_i' \alpha + g(w_i)), \quad (51)$$

where $\Lambda(\cdot)$ is the logistic function and $g(\cdot)$ is a continuous piecewise-linear function of the log wage offer. The no-offer control group is assigned a propensity score of zero by design. Let $\kappa_0 < \kappa_1 < \dots < \kappa_R$ denote knots spanning the support of w_i . I write the wage-offer component as

$$g(x) = \sum_{r=1}^R \exp(\delta_r) h_r(x), \quad (52)$$

where $h_r(x) \equiv [\min\{x, \kappa_r\} - \kappa_{r-1}]_+$ is the cumulative length of interval $[\kappa_{r-1}, \kappa_r]$ reached by x , with $[a]_+ \equiv \max\{a, 0\}$. Equivalently, $g(x)$ is linear within each knot interval, with slope $\exp(\delta_r)$ on interval r . Since $\exp(\delta_r) > 0$, the fitted first-stage index is monotone increasing in the log wage offer holding X_i fixed. Because the logit link is also monotone, the fitted propensity score is likewise monotone increasing in the offered wage.

The parameters are estimated by penalized logit maximum likelihood:

$$\left(\hat{\alpha}_0, \hat{\alpha}, \hat{\delta}\right) = \underset{\alpha_0, \alpha, \delta}{\operatorname{argmin}} \sum_i \omega_i [\log \{1 + \exp(\eta_i)\} - D_i \eta_i] + \frac{\lambda}{2} \alpha' \alpha, \quad (53)$$

where $\eta_i = \alpha_0 + X_i' \alpha + g(w_i)$ and ω_i denotes any estimation weight. The ridge penalty is applied only to the linear covariate coefficients in the first stage, not to the wage-offer spline. This regularization stabilizes specifications with richer control sets while preserving full monotonicity and flexibility in the wage-offer dimension. The implementation uses five interior knots, placed at equally spaced quantiles of log wage offers unless otherwise specified, and sets $\lambda = 25$. The second-stage MTE estimation is otherwise held fixed at the baseline semi-parametric specification, including the second-degree local polynomial and bandwidth of 0.2.

Appendix F Derivations of MVPFs and Optimal Policies

In this appendix, I derive optimal tax and subsidy policies from Section 6.

Subsidy

Consider an hourly-wage subsidy of δ per hour worked. In my model, the subsidy lowers nominal reservation wages by δ , yielding a new equilibrium share of hourly workers, θ^δ , such that

$$\bar{w}(\theta^\delta) = AV_1(\theta^\delta) + \delta. \quad (54)$$

The aggregate willingness-to-pay for the subsidy is

$$WTP(\delta) = \underbrace{\delta\theta^\delta}_{\text{Transfer}} + \underbrace{\int_{\theta^{EQ}}^{\theta^\delta} (MV_1(\theta) - \bar{w}(\theta)) d\theta}_{\text{Insurance Benefit}}, \quad (55)$$

and the government's net cost is

$$NC(\delta) = \underbrace{\delta\theta^\delta}_{\text{Direct Cost of Transfer}} - \underbrace{\int_{\theta^{EQ}}^{\theta^\delta} \tau MH(\theta) d\theta}_{\text{Fiscal Externality from Moral Hazard}}. \quad (56)$$

The corresponding MVPF is

$$MVPF_{\text{Sub}}(\delta) \equiv \frac{WTP(\delta)}{NC(\delta)} = \frac{\delta\theta^\delta + \int_{\theta^{EQ}}^{\theta^\delta} (MV_1(\theta) - \bar{w}(\theta)) d\theta}{\delta\theta^\delta - \int_{\theta^{EQ}}^{\theta^\delta} \tau MH(\theta) d\theta}. \quad (57)$$

The welfare-maximizing level of subsidy solves

$$\max_{\delta} \left\{ \underbrace{\delta\theta^\delta + \int_{\theta^{EQ}}^{\theta^\delta} (MV_1(\theta) - \bar{w}(\theta)) d\theta}_{WTP(\delta)} - \lambda \underbrace{\left(\delta\theta^\delta - \int_{\theta^{EQ}}^{\theta^\delta} \tau MH(\theta) d\theta \right)}_{NC(\delta)} \right\}, \quad (58)$$

where λ reflects the marginal cost of public financing—the cost of raising one dollar of revenue through taxation, or the MVPF of some alternative policy from which funds

are redirected. The first order conditions for (58) imply

$$MVPF_{dSub}(\delta^*) \equiv \frac{WTP'(\delta^*)}{NC'(\delta^*)} = \frac{\frac{d\delta}{d\theta^\delta}\theta^{\delta^*} + \delta^* + MV_1(\theta^{\delta^*}) - \bar{w}(\theta^{\delta^*})}{\frac{d\delta}{d\theta^\delta}\theta^{\delta^*} + \delta^* - \tau MH(\theta^{\delta^*})} = \lambda. \quad (59)$$

$MVPF_{dSub}(\delta)$ is the MVPF for a *marginal increase* in hourly-wage subsidy.

To calculate $\frac{d\delta}{d\theta^\delta}$, consider the equilibrium condition from Equation (11) in the presence of an hourly-wage subsidy, δ :

$$\bar{w}(\theta^\delta) = AV_1(\theta^\delta) + \delta. \quad (60)$$

Differentiating with respect to θ^δ yields

$$\frac{d\delta}{d\theta^\delta} = \frac{d\bar{w}}{d\theta^\delta} - \frac{dAV_1(\theta^\delta)}{d\theta^\delta} \quad (61)$$

$$= \left(\frac{dS}{d\bar{w}}\right)^{-1} - \frac{MV_1(\theta^\delta) - AV_1(\theta^\delta)}{\theta^\delta} \quad (62)$$

$$= \frac{\bar{w}}{\beta\theta^\delta(1 - \theta^\delta)} - \frac{MV_1(\theta^\delta) - AV_1(\theta^\delta)}{\theta^\delta} \quad (63)$$

$$(64)$$

where β is the coefficient on a $\ln \bar{w}$ in a logistic model of hourly labor supply.

Tax

The welfare-maximizing level of tax is given by

$$\max_{\rho} \{ \eta NR(\rho) - WTP(\rho) \}, \quad (65)$$

where $NR(\rho)$ is the government's revenue from the piece-rate tax net of any fiscal externalities, and $WTP(\rho)$ is aggregate private willingness-to-pay to *avoid* the tax. η reflects the highest-MVPF policy for which revenue might be used. Under a balanced budget, η would represent the MCPF of the least efficient revenue source one could replace with the piece-rate tax.

$$\max_{\rho} \left\{ \eta \left(\underbrace{\int_{\theta^{\rho}}^1 \rho MV_0(\theta) d\theta + \int_{\theta^{EQ}}^{\theta^{\rho}} \tau MH(\theta) d\theta}_{NR(\rho)} - \underbrace{\left(\int_{\theta^{\rho}}^1 \rho MV_0(\theta) d\theta - \int_{\theta^{EQ}}^{\theta^{\rho}} (MV_1(\theta) - \bar{w}(\theta)) d\theta \right)}_{WTP(\rho)} \right) \right\}, \quad (66)$$

The first order conditions for (66) imply

$$MCPF_{dTax}(\rho^*) \equiv \frac{WTP'(\rho^*)}{NR'(\rho^*)} = \eta \quad (67)$$

$$\frac{\frac{d\rho}{d\theta^{\rho}} \int_{\theta^{\rho^*}}^1 MV_0(\theta) d\theta - \rho^* MV_0(\theta^{\rho^*}) - (MV_1(\theta^{\rho^*}) - \bar{w}(\theta^{\rho^*}))}{\frac{d\rho}{d\theta^{\rho}} \int_{\theta^{\rho^*}}^1 MV_0(\theta) d\theta - \rho^* MV_0(\theta^{\rho^*}) + \tau MH(\theta^{\rho^*})} = \eta. \quad (68)$$

$MCPF_{dTax}(\rho)$ is the MCPF for a *marginal increase* in piece-rate tax.

To calculate $\frac{d\rho}{d\theta^{\rho}}$, consider the equilibrium condition from Equation (11) under the tax wedge defined in Section 6:

$$\bar{w}(\theta^{\rho}) = (1 + \rho) AV_1(\theta^{\rho}). \quad (69)$$

Differentiating with respect to θ^{ρ} yields

$$\frac{d\rho}{d\theta^{\rho}} = \frac{\frac{d\bar{w}}{d\theta^{\rho}}}{AV_1(\theta^{\rho})} - \frac{\bar{w}(\theta^{\rho})}{AV_1(\theta^{\rho})^2} \frac{dAV_1(\theta^{\rho})}{d\theta^{\rho}} \quad (70)$$

$$= \left(\frac{dS}{d\bar{w}} \right)^{-1} \frac{1}{AV_1(\theta^{\rho})} - \frac{\bar{w}(\theta^{\rho})}{AV_1(\theta^{\rho})^2} \frac{MV_1(\theta^{\rho}) - AV_1(\theta^{\rho})}{\theta^{\rho}} \quad (71)$$

$$= \frac{\bar{w}(\theta^{\rho})}{\beta \theta^{\rho} (1 - \theta^{\rho}) AV_1(\theta^{\rho})} - \frac{\bar{w}(\theta^{\rho}) (MV_1(\theta^{\rho}) - AV_1(\theta^{\rho}))}{\theta^{\rho} AV_1(\theta^{\rho})^2} \quad (72)$$

where β is the coefficient on a $\ln \bar{w}$ in a logistic model of hourly labor supply.

General Equilibrium Effects The first-order effect of an hourly-wage subsidy or piece-rate tax is to increase the firm's relative value of hiring fixed-wage labor, corresponding to an upward shift in the average value curve. In equilibrium, however, we would also expect both policies to lower the effective piece rate paid to non-fixed-wage workers. Because the piece rate, p , represents hourly workers' outside option, this decrease would likely result in a downward shift in hourly reservation wages,

$\bar{w}(\theta)$. The result would be to further increase the equilibrium share of hourly workers, magnifying the welfare effects of the policies above.⁵⁷ In other words, omitting these second-order effects from my analysis understates the social efficiency of wage subsidies and piece-rate taxes, biasing my estimates of $MVPF(\delta)$ and $MVPF(\rho)$ towards one.

Relatedly, the above analysis assumes that aggregate demand for labor product, q , is perfectly inelastic. Relaxing this assumption would mean both hourly wage subsidies and piece-rate taxes could generate less social value by distorting the equilibrium quantity of labor product away from its efficient level. Note, however, that because these distortions move in opposite directions, one might mitigate their potential welfare impacts with a mix of taxes and subsidies.

⁵⁷Appendix Figure A4 provides a graphical illustration of both first- and second-order effects of hourly-wage subsidies or piece-rate taxes on hourly labor share. Incorporating the second-order effect into my welfare analysis would require estimates of firms' demand for labor product, q , and the elasticity of hourly labor supply with respect to piece-rates, p .

Appendix G External Validity

My experiment was designed to demonstrate how moral hazard and adverse selection can lead to an underprovision of fixed-wage jobs. In the parlance of List (2020), it aims to provide “WAVE1” insights validating the theory presented in Section 4. While point estimates directly speak to the importance of information asymmetries in some settings, they are not intended to measure welfare losses across the myriad of labor markets characterized by risky forms of compensation. Nonetheless, it is important to consider the settings to which my estimates can be directly applied. I therefore assess the external validity of my results using the SANS conditions from List (2020).

Selection The experimental sample was drawn from the population of workers on Prolific, a widely used and well-established freelancing platform with over 100,000 workers. Among online workers, Prolific is widely considered the most desirable micro-task platform due to its ease of use and high pay. Consequently, the platform has a waitlist, and workers rarely turn down a task for which they are eligible (Douglas et al., 2023). Workers in this population were eligible for my experimental job posting if they met four screening criteria: (1) were located in the United States, (2) spoke fluent English, (3) successfully completed ten or more previous tasks, and (4) earned an approval rate above 98 percent on previous tasks. More than 95 percent of Prolific workers meet the 98-percent approval threshold. Importantly, my experimental job posting advertised generous compensation for a five-minute task—a \$1.00 flat fee in addition to the \$0.03 piece rate for each correctly typed sentence—ensuring the sample is not restricted to workers with low reservation wages or limited outside options. Experimental jobs were posted across a broad range of days and times to ensure the sample was not restricted to workers with a particular schedule.

Attrition There was virtually no attrition from the sample. One worker exited the task before observing their experimental wage offer and was dropped from the experimental sample. All other workers remained in the sample, even if they failed to enter sentences or click the submit button after the five-minute timer expired.

Naturalness Each aspect of the experimental intervention was designed to place participants in a naturally-occurring setting. Contract options are presented in a simple and straightforward manner, offering workers a choice between a randomized

hourly wage offer and a standardized \$0.03 piece rate for each correctly typed sentence. This \$0.03 piece rate was set to approximate observed rates for online text-to-text transcription services (Khan, 2024; Ahmad, 2024; GMR Transcription, 2024; GoTranscript, 2024; Ditto Transcripts, 2024; Transcription Services, 2024). Like most online employment, hired workers faced a threat of job dismissal and reputational damage for unapproved tasks, so they had an incentive to maintain a minimum standard of performance under either compensation scheme. The transcription task, described in Appendix C, is commonly requested by clients on online platforms (Khan, 2024; Ahmad, 2024), and “traditional keyboarding” is a job requirement for 66 percent of American workers (Bureau of Labor Statistics, 2024). Finally, participants were not aware that they were part of an experiment until after performing the task, so estimates are not biased by any desire to generate a particular result.

Scaling My experimental task is more predictable, lower stakes, and shorter duration than many of those found in other labor markets. As a result, workers might exhibit less fatigue, more inattention, and a greater tolerance for risk than those in other work settings. I would expect these attributes to attenuate my estimates towards zero: Workers facing higher stakes or more uncertainty over their labor products would pay a higher premium (lower reservation wage) for the implicit insurance offered by fixed wages.⁵⁸ This increased risk premium would shift the hourly supply curve downwards, resulting in a greater welfare loss than the one I estimate. Unless risk aversion is high enough sustain fixed wages for the entire market, higher stakes (i.e., more risk aversion) would typically result in a greater welfare loss. In other words, my analysis should *underestimate* the welfare loss we would expect from output-based compensation with higher stakes than those in my experiment. Likewise, less inattention would make hourly supply more elastic and more correlated with workers’ latent productivity, exacerbating adverse selection problems. Finally, more fatigue would likely lead to larger moral hazard effects—if the cost of effort increases with the duration of the task, so would the benefits of shirking.⁵⁹

⁵⁸Workers’ uncertainty over their labor product depends on their prior knowledge of the task. To be conservative, I intentionally design the task and instructions to maximize this task-related knowledge while maintaining realism. If instructions were less informative, each worker would face a higher subjective variance in earnings and a larger benefit from insurance.

⁵⁹In light of these potentially attenuating forces, the fact that workers in my experiment still make strategic, risk-averse decisions is noteworthy. For example, despite facing moderately predictable, short-term task with small monetary stakes, the majority of workers produce output values above their reservation wages.

While my welfare estimates are likely attenuated relative to many short-term labor markets, this lower-bound relationship is unlikely to hold in every setting, especially those with dynamically adjusting contracts or costly output monitoring.⁶⁰ Policies aimed at addressing information asymmetries in these markets warrant further evidence targeted to those settings.

⁶⁰I discuss how my framework can be adapted to accommodate labor-market features like monitoring costs and dynamic contracts in Appendix D.

Appendix H Selection into Lease-Free Rideshare Driving: Calibrations from Angrist et al. (2021)

In Angrist et al. (2021), experimental “taxi” offers consist of a reduced proportional fee on fares charged to drivers, t_1 , and a weekly lease payment, L . For a driver who earns Y_i per week in ride fares, accepting an offer would increase their take-home pay from $(1 - t_0)Y_i$ to $(1 - t_1)Y_i$, where t_0 is the pre-existing Uber fee of 20 or 25 percent, in exchange for paying Uber $\$L$ each week. For a given lease offer (t_1, L) , the “breakeven value,” w_B , is defined as the weekly fare revenue required for a driver’s increase in take-home earnings, $(t_0 - t_1)Y_i$ to cover the cost of their weekly lease payment, L :

$$w_B \equiv \frac{L}{\Delta_t}, \quad (73)$$

where $\Delta_t \equiv t_0 - t_1$. So, a lease offer (L, t_1) , effectively gives the driver an opportunity to buy back a Δ_t -share of their own earnings at a valuation or “share price” of w_B . Viewed through this lens, opting into a taxi-style lease is the same as opting *out* of a (partial) fixed wage contract—a driver who accepts a lease offer gives up $\$L$ in fixed weekly income in favor of a Δ_t -higher share of their labor product. This equivalence means I can calibrate my model using estimates of selection *out of* taxi leases from Angrist et al. (2021) to approximate welfare losses from inefficient compensation in the market for rideshare drivers.

To calibrate my model to this setting, I estimate a supply curve for lease-free driving by fitting a logit model to the aggregated statistics from Table 4 of Angrist et al. (2021), which reports lease take-up rates and observation counts by week, experimental strata, and treatment condition. Specifically, I derive $S(w_B) \equiv \Pr[1 - Lease_i | w_{Bi} = w_B]$ by estimating the following:

$$\log \left(\frac{1 - Lease_i}{Lease_i} \right) = \alpha + \beta \ln w_{Bi} + \gamma \mathbf{X}_i, \quad (74)$$

where $\mathbf{X}_i$ includes dummy controls for experimental strata and week. To derive average value curve, I combine the information on take-up rates with reported realized gains across accepters and decliners of leases. First, I use values in Table 4 to compute

average experimental breakeven offers, residualized by strata- and week-fixed effects, separately for accepters and decliners of lease offers. I then combine these estimates with information from Table 6, which reports average gains above breakeven offers and share driving more than zero hours separately by accepters and decliners of experimental offers. From Table 4, the average residualized breakeven value among decliners is \$274. From Table 6, the share of decliners who drove more than zero hours is $423/679 = 62\%$. Among those who drove, accepters gained an average of 115. So the average earnings among decliners is $(\$115 + \$274) \times 0.62 = \$242$. Because this value is computed among decliners in all experimental groups,⁶¹ I let it correspond to the average fare revenue among the 55 percent drivers who opt out across all lease offers after controlling for strata and week, so

$$AV(0.55) \equiv E[Y_i|\theta \leq 0.55] = \$242. \quad (75)$$

Meanwhile, the average residualized breakeven value among accepters in Table 4 is \$271. From Table 6, the share of accepters who drove more than zero hours is $515/560 = 92\%$. Among those who drove, accepters gained an average of 97. So the average earnings among accepters is $(\$97 + \$271) \times 0.92 = \$338$. This value corresponds to the average fare revenue among the 45 percent drivers who opt in across all lease offers after controlling for strata and week, so $E[Y_i|\theta > 0.55] = \$338$.⁶² I combine this value with $AV(0.55)$ from Equation (75) to take the total expectation,

$$\begin{aligned} AV(1) &\equiv E[Y_i|\theta \leq 1] = E[Y_i] \\ &= 0.55 \times E[Y_i|\theta \leq 0.55] + (1 - 0.55) \times E[Y_i|\theta > 0.55] \\ &= 0.55 \times \$242 + (1 - 0.55) \times \$338 \\ AV(1) &= \$285. \end{aligned} \quad (76)$$

I therefore calibrate the average value curve by fitting a line between two points:

⁶¹Unfortunately, Angrist et al. (2021) does not report statistics by take-up status at the strata, week, or breakeven level, so I must infer the average value curve from acceptor and decliner averages across all treatment conditions. As a result, the average value curve is calibrated using just two points—one on either side of the average log wage offer and its corresponding supply share. Note that this aggregation likely attenuates the estimated slope of $AV(\theta)$, resulting in smaller estimates of welfare loss.

⁶²Note that Table 6 values in Angrist et al. (2021) are already adjusted to remove the labor-supply effects of leases. As such value-curve estimates correspond to $AV_1(\theta)$ and $MV_1(\theta)$ —potential output under the partial fixed wage contract.

$AV(0.55) = \$242$ and $AV(1) = \$285$, yielding the following average value curve:

$$AV(\theta) = 189 + 96\theta. \quad (77)$$

With a linear estimate of the average value curve in hand, the marginal value curve can be computed as a simple derivative:

$$\begin{aligned} MV(\theta) &= \frac{\partial}{\partial \theta} (\theta AV(\theta)) \\ &= \frac{\partial}{\partial \theta} (189\theta + 96\theta^2) \\ MV(\theta) &= 189 + 192\theta. \end{aligned} \quad (78)$$

With average value, marginal value, and supply curves in hand, I can calculate equilibrium and efficient shares of lease-free driving. The equilibrium share of lease-free contracts is $\theta^{EQ} = 0.55$. The associated “breakeven price” of $w_B(\theta^{EQ}) = \$242$ represents the breakeven value on the hypothetical lease that drivers would need to pay in lieu of the proportional fee on rider fares to generate the same net revenue for Uber. Meanwhile, the efficient share of lease-free contracts we would expect in a full-information equilibrium is $\theta^{EF} = 0.59$. The difference of $\theta^{EF} - \theta^{EQ} = 0.05$ represents potential drivers who would generate profitable levels of fare revenue if they were charged a lower proportional fee or paid some additional fixed weekly compensation that implicitly valued their labor product somewhere between \$242 and \$303 per week.⁶³ However, the cost of adverse selection among less productive drivers would make offering this compensation unprofitable. The corresponding welfare loss, shaded in yellow, is \$1.32 per week. Table 3 of Angrist et al. (2021) reports the average driving hours per week as 14.16, so the deadweight loss is $\$1.32/14.16 = \0.09 per hour worked, comparable to my estimates of welfare loss in the market for online data-entry.⁶⁴ Appendix Figure A7 plots the MCPF of a tax on drivers’ fare earnings. Taxes of 23% or less on driver fares generate MCPFs between .86 and .95—better

⁶³One way to frame this comparison is to imagine that all drivers must pay a fixed weekly “lease,” $\bar{L}$, to operate a taxi or Uber. In equilibrium, the θ^{EQ} -share of drivers who choose Uber forfeit Δ_t of their earnings in exchange for a fixed weekly wage of $w_B^{EQ} \Delta_t = \bar{L}$, exactly offsetting the lease. A higher breakeven value, $w_B > w_B^{EQ}$, could be accomplished with a lower proportional fee, $w_B = \frac{w}{t_0 - \delta} > w_B^{EQ}$, or with a higher fixed wage, $w_B = \frac{w + \epsilon}{t_0} > w_B^{EQ}$ (i.e., drivers receive a fixed $\epsilon = w_B - \bar{L}$ per week).

⁶⁴Note that, because this exercise considers mixed contracts, I exclude moral hazard from the welfare analysis. In effect, I am identifying the insurance value drivers’ place on contracts with marginally lower lease payments (L) and marginally higher Uber fees (t). As such, my estimates represent the welfare losses from adverse selection only.

than a distortionless tax.

There are several limitations to this calibration exercise. Calibrated curves, particularly the average value curve, rely on transformations across a small number of point estimates. Linear extrapolating market-wide selection patterns from these points comes with strong functional-form assumptions, and the absence of standard errors prevents any meaningful discussion of statistical significance. Moreover, the experimental sample in Angrist et al. (2021) consists of *existing* Uber drivers. Observed selection into taxi-style lease offers likely excludes drivers with the strongest preference for this type of compensation because they are already working as taxi drivers! More generally, studies of workers under existing employment contracts are likely to understate the effects of information asymmetries, which depend upon marginal treatment and selection across all workers in a hypothetical market. Finally, while drivers in Angrist et al. (2021) are stratified into high- and low-hour groups with separate experimental offers, this coarse conditioning likely falls short of the kind of dynamic contract that’s feasible on modern rideshare platforms. For example, the “Uber Pro” program gives drivers with favorable histories access to more favorable rates and amenities, potentially mitigating welfare loss from adverse selection (Uber Technologies Inc., 2025). Despite these limitations, however, the calibration exercise provides an example of how my framework can be applied to a variety of labor markets where workers selection on potential productivity can prevent efficient contracting.